The Architectural Cost of Hybrid VTOL: meryemAircraft, a Propeller-Driven Tail-Sitting Blended-Wing-Body Without a Dedicated Lift System

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Abstract. Hybrid vertical take-off and landing (VTOL) aircraft combine runway independence with wing-borne cruise and pay for it in cruise efficiency. This paper treats that cost as architectural rather than as a defect of implementation, and develops it as an accounting framework. The penalty is charged in three coupled currencies — the mass of hover hardware carried through cruise, its drag when exposed in cruise, and a power system sized by a condition holding for some two percent of the flight — and each of the remedies surveyed here reduces one by raising another. The escape condition is then explicit, and it has four parts: the penalty is charged unless hover and cruise are served by the same hardware, in the same orientation, doing the same job, with the hover peak supplied from a buffer rather than from permanently installed continuous power. A second result is methodological: architectural comparisons depend on the sizing contract chosen, and a fixed fuel fraction removes the mass bill from the range column altogether, so three contracts are reported rather than one. meryemAircraft, an uncrewed tail-sitting blended-wing body, satisfies the escape condition and serves as the case study: one coaxial pair at the nose gives all thrust in both regimes, four small pairs at the tips give attitude moments only, and a deployable strip is assigned the roll that body-parallel thrust cannot produce. Sized against a lift-plus-cruise layout on wind-tunnel drag, it closes the same mission at forty-two percent lower take-off mass and seventeen percent greater range; against a tilting layout the comparison reverses between contracts and no superiority is claimed. Those are sizing results for an aircraft that has not been built: a three-dimensional solution bounds the zero-lift drag with a quantified sensitivity budget, but the component mass build-up closes the 50 kg design only on a battery specific power of 5.6 kW kg^{-1}, which is 3.8 times the highest rate yet measured on a flown pack, and it does not close the 1000 kg design at all. The study is analytical, with no experimental validation of the configuration. The tip propellers can turn the aircraft's rotational inertia through the transition but not, on present evidence, its aerodynamic moment. Resolving that margin along the trajectory shows the aircraft never reaches ninety degrees of incidence — the relative wind rotates with the body — so the outstanding measurement is the pitching moment to some twenty-two degrees at low dynamic pressure, together with trim at cruise. A vortex-lattice solution establishes static pitch stability under a stated packaging rule and closes cruise trim, though not in the way first supposed: measured data for reflexed sections fall an order of magnitude short of the moment required, while nine degrees of tip washout supplies it, at a cost of 4.3 percent of cruise lift-to-drag ratio. Roll is treated the same way: the inertia and the damping are computed for this planform, the

moment needed for a twenty-degree-per-second roll follows from them, and the strip's effectiveness in supplying it is stated as a requirement rather than demonstrated. Yaw has the strongest authority of the three axes, because differential tip thrust acts through the semi-span, but the planform supplies no directional stability at all, so the tip-frame fairings must serve as the vertical surfaces as well as the drag measure they were introduced as. Attitude control is therefore sized in every axis and closed in none. Transition controllability remains the principal open requirement and is stated as a threshold a future measurement must meet.

Keywords: vertical take-off and landing; tail-sitter; blended wing body; uncrewed aerial vehicle; series hybrid propulsion; cruise efficiency; aircraft configuration design

1. Introduction

Powered flight for uncrewed aircraft is dominated by two configuration families, and each is bounded by a different limit.

Fixed-wing aircraft carry payload efficiently over long distances because the wing sustains the vehicle without continuously expending power on lift. Their limit is not aerodynamic but infrastructural: they require a runway, a catapult, or an equivalent launch and recovery installation. That requirement is expensive, fixed in place, and scales poorly — a larger fixed-wing aircraft demands a longer runway, stronger pavement, and wider taxiways, so its growth is gated by the ground rather than by the air.

Rotary-wing and multirotor aircraft remove that requirement entirely. They take off and land vertically, hover, and operate from confined sites. Their limit is the converse: without a wing, every second of flight is paid for with installed power, so range and endurance remain modest and degrade further as the vehicle grows. Figure 1 places the two families against the two capabilities and marks the corner that neither occupies.

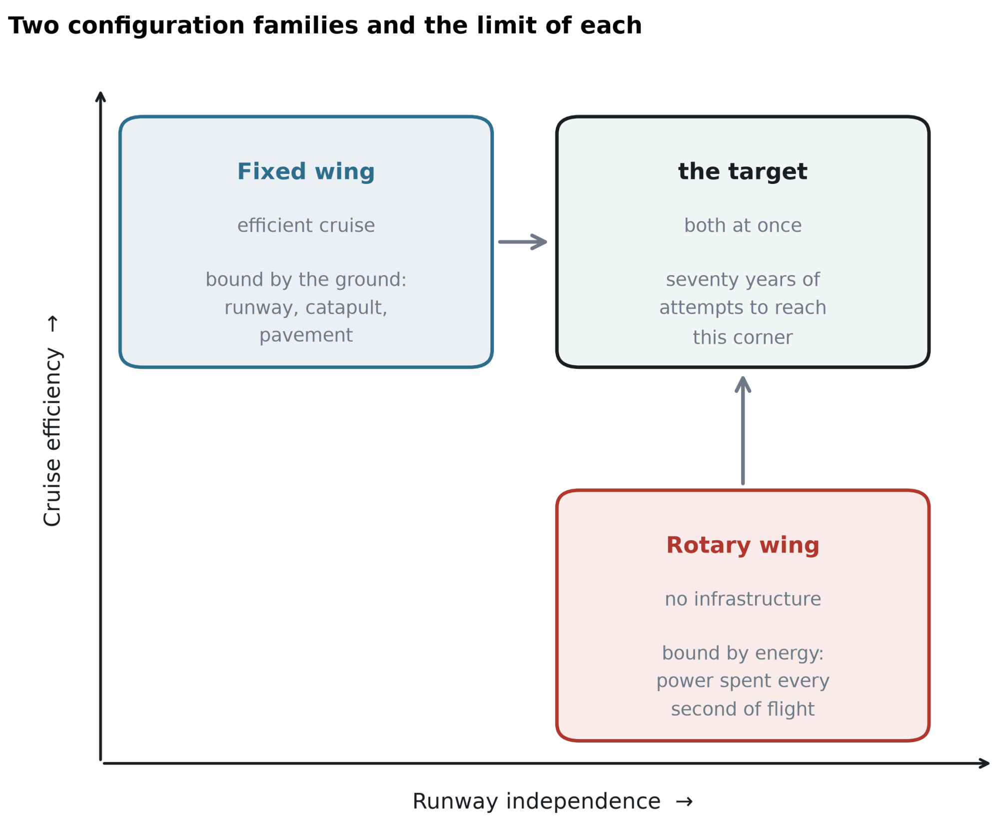


Figure 1. Two configuration families and the limit of each. Fixed-wing aircraft occupy the efficient-cruise corner and are bounded by ground infrastructure; rotary-wing aircraft occupy the runway-independent corner and are bounded by installed power. Seventy years of hybrid development has been an attempt to reach the corner that neither family occupies.

The demand to combine the two has been continuous and expensive. Tail-sitting prototypes were flown in the 1950s, vectored-thrust and tilt-wing aircraft in the 1960s, tilt-rotors from the 1980s, and a broad family of hybrid vertical take-off and landing (VTOL) uncrewed aircraft since the 2010s. Different nations, services and propulsion philosophies have attacked the same problem for seventy years. No field sustains that level of effort against a need that is not real.

Contemporary hybrid VTOL aircraft do combine the two capabilities, and several are in service. This paper does not dispute that. It argues, instead, that they purchase the capability at a specific and quantifiable price, and that the price is charged to cruise efficiency.

The most common hybrid architecture — separate lifting rotors for the vertical phase and a separate propulsor for cruise — carries its hover hardware for the whole flight while using it for a few percent of it. The consequences are measurable. Lift propellers left exposed in forward flight impose a parasitic drag penalty large enough that retracting them recovers a substantial fraction of it; rotor-wake interference raises drag further in the hybrid regime; and the second propulsion system remains as dead mass for the entire cruise. Tilting architectures avoid the dead mass but substitute mechanical complexity, gyroscopic coupling during transition, and a non-trivial transition control problem.

The central observation of this paper is that these penalties are not defects of implementation. They arise from the architecture itself: from the decision to provide hover and cruise with different hardware. Better engineering can move the penalty between mass, drag and powertrain sizing, and can reduce any one of them, but it cannot remove the trade, because the trade is structural.

This paper presents a configuration that does not make that trade. In the proposed arrangement the aircraft sits on its tail, its entire body is a lifting blended-wing body, thrust for every flight phase is produced by a single coaxial counter-rotating propeller pair at the nose, and attitude control is produced by four small coaxial pairs mounted on rigid frames at the wing tips. The aircraft has no elevons, no rudder, no tilting mechanism and no dedicated lift system. Roll authority, which cannot be generated by thrust vectors parallel to the body axis, is assigned to a level-controlled deployable strip on the lower surface positioned within the main propeller slipstream, so that it is loaded at zero airspeed. Section 4.4 computes the roll inertia and the roll damping of this planform and states the moment the strip must supply; it does not show that the strip supplies it.

Because the same primary propulsor serves hover and cruise without changing its orientation relative to the airframe, none of the three penalties **as defined in Section 3** arises in full: there is no second thrust system to carry, no lift hardware left exposed in the cruise airstream, and no *engine* sized by the hover peak. The third of those is the one that comes with a qualification, and Section 5.3 states it rather than deferring it: the buffered series hybrid releases the engine and its fuel consumption from the hover condition, but the electric machines and the power electronics still pass the full hover power, and the hover-sized machine is the largest single item in the propulsion chain. That is a statement about three specific charges, two of which do not arise and one of which is halved, not a claim that the configuration is free. What it pays instead — the mass and drag of the control propellers and their supporting frames, the rolling-moment device, and the transition manoeuvre itself — is reported and quantified in Section 5.4 rather than omitted.

Contributions. The primary contribution is a framework; the aircraft is the case that instantiates it. The paper

1. **states the cruise-efficiency penalty of hybrid VTOL as an architectural property** rather than a defect of implementation, expresses it as three dimensionless charges — carried hover mass, exposed cruise drag, and continuous power sized by the hover peak — shows with published figures that the known remedies transfer the penalty between them rather than removing it, derives an explicit escape condition, and tests a consequence against an independent published sizing study: that the architecture with the best cruise efficiency need not be the lightest, which is what that study reports and what a single-metric comparison would not anticipate;
2. **shows that architectural comparisons are contract-dependent**, a methodological result independent of any particular aircraft: range computed at a fixed fuel fraction is independent of take-off mass, so the mass bill never reaches the range column. Three sizing contracts are reported side by side and the ranking changes between them;
3. **instantiates the escape condition in a configuration**, audits the three bills against it one at a time, and states what the configuration pays instead;

4. **supports the case study with computation rather than assertion** where it could — a three-dimensional Reynolds-averaged solution for the zero-lift drag with a quantified sensitivity budget across grids, wall resolutions, turbulence closures and starting fields; a component mass build-up that closes the 50 kg design conditionally and does not close the 1000 kg one; a rotational check establishing inertial feasibility of the transition; and a viscous, station-by-station solution of the trimmed wing that corrects the assumed span efficiency downward; and
5. **states what is not established, as a testable requirement rather than an omission.** The aerodynamic pitching moment through the rotation is not known, and the paper reports the coefficient that would consume the available control margin instead of estimating the coefficient itself.

Items 1 and 2 stand independently of whether this aircraft is ever built. Item 5 is the reason the paper does not claim that it can be.

Scope. This is a configuration study containing no wind-tunnel measurement and no flight test. Its results are analytical estimates from stated assumptions, with two exceptions computed here: the zero-lift drag of the wing and centre body, solved three-dimensionally, and the span efficiency of the trimmed wing, solved station by station with a viscous section method. The mass budget began as a target rather than a finding; a component build-up replaces it for the light design, closes conditionally, names the condition, and does not close the heavy design at all. Section 8 states these limitations, and Supplementary S5 enumerates all of them.

The remainder of the paper is organised as follows. Section 2 reviews seventy years of attempts to merge the two configuration families and argues, on the evidence of two contemporary NASA reviews, that they ended for reasons external to the configuration — principally engine and transmission reliability — while the one difficulty those reviews document most fully, the workload of a pilot flying a vertical descent, is also the one that an uncrewed aircraft removes outright. Section 3 sets out the architectural tax in its three currencies, shows that architectural remedies transfer it rather than remove it, and derives the condition under which it would not be charged. Section 4 describes the proposed configuration, which is built to satisfy that condition. Section 5 audits the claim bill by bill and states what the configuration does pay. Section 6 sizes two reference designs twenty times apart in mass and examines how the proportions scale. Section 7 treats the flight profile and the transition manoeuvre, including a result that contradicts a common assumption about how quickly a tail-sitter should rotate. Section 8 states the limitations, and Section 9 concludes.

2. Background: seventy years of attempts

The configuration proposed in this paper is new, but the problem it addresses is not, and neither are several of its ingredients. This section reviews the attempts that preceded it. The purpose is not to establish priority but to establish two things: that the need has been pursued continuously for seventy years, and that the pursuit was rarely abandoned because the aerodynamics failed. Figure 2 places the programmes discussed below on a single timeline, with the recorded reason each one stopped.

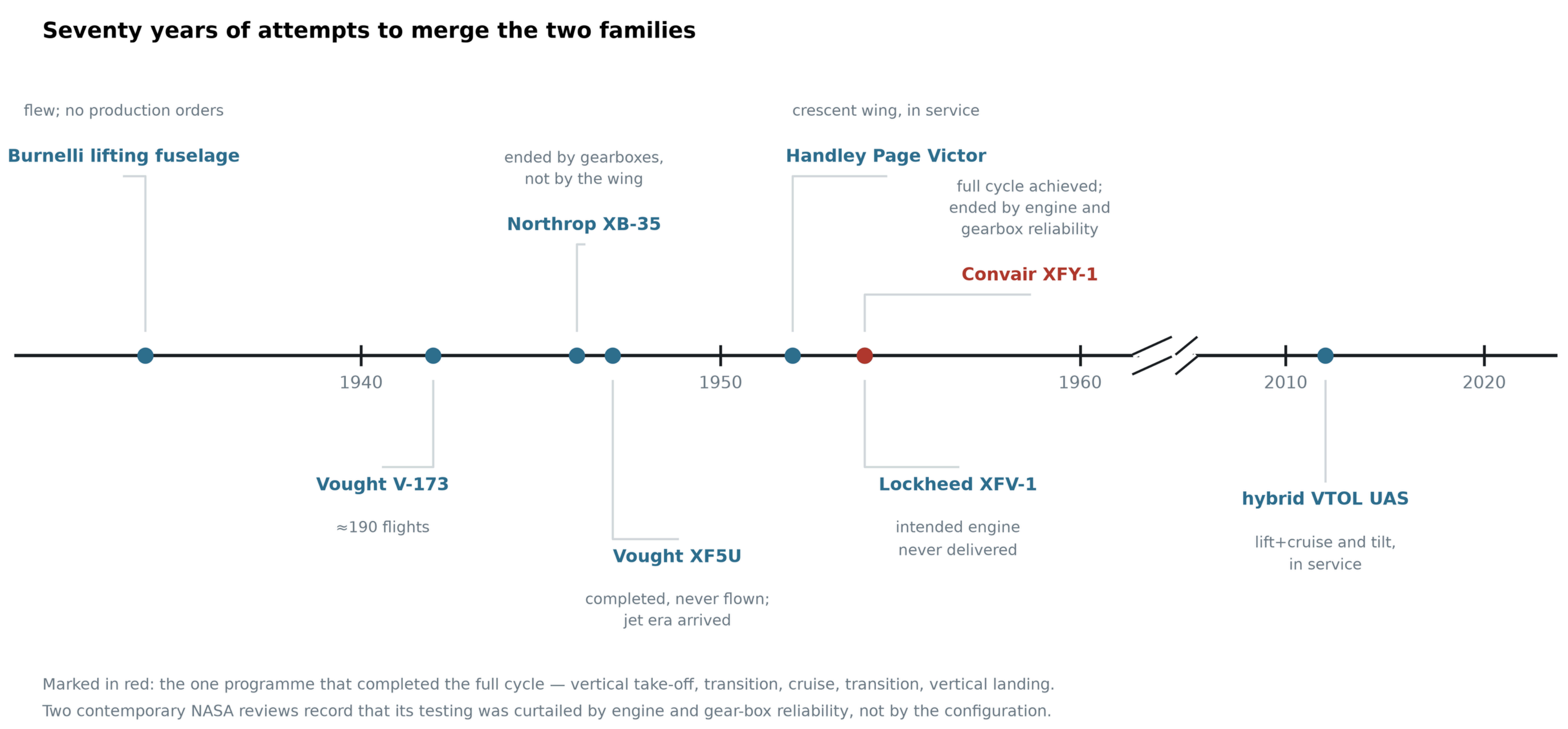


Figure 2. Seventy years of attempts to merge the two families, with the recorded reason each programme ended. The two entries of 1954 and the reason the XFY-1 programme stopped are taken from [1] and [2]; the remaining entries carry no numerical claim.

2.1 Removing the fuselage

The idea that a transport aircraft should carry its payload inside a lifting surface rather than inside a cylinder is as old as the transport aircraft itself. Burnelli's lifting-fuselage designs, flown in a succession of prototypes from the 1920s to the 1940s, placed cabin and cargo inside a thick aerofoil-shaped centre body that contributed lift instead of only drag. The aircraft flew, repeatedly and over two decades. They never entered series production. The reasons for that are outside the scope of this paper and are disputed; what matters here is only that the layout was flown rather than merely proposed, and was not abandoned at the drawing board.

The Northrop XB-35 carried the same idea to its limit: a bomber with no fuselage and no tail at all. Its programme is often cited as evidence that the flying wing was premature. Whatever weight that reading deserves, the aircraft's best-documented difficulties lay in its power transmission rather than in its aerodynamics: the contra-rotating propellers were driven through remote gearboxes and long extension shafts, and were eventually changed to single-rotation units. The distinction between an airframe and the machinery installed in it matters for the present work, and Section 4.3 returns to it — the configuration proposed here uses counter-rotating propellers but never builds a contra-rotating gearbox, because each rotor of a pair is driven by its own electric machine on a common axis.

Vought's V-173 and XF5U pursued the opposite extreme of the same intuition — a wing of very low aspect ratio with large propellers at the tips, intended to work against the tip vortices. The V-173 flew roughly two hundred times and supported the low-speed claims made for it. The XF5U was completed and never flown; the programme ended as the services moved to jet propulsion.

The recurring pattern is worth stating plainly: in each case the aircraft flew, the configuration was not disqualified in flight, and the programme ended for a reason that came from outside the configuration.

2.2 Standing the aircraft on its tail

The tail-sitter is the most direct answer to the runway: if the aircraft points its thrust line at the ground it needs no separate lift system, no tilting mechanism and no second propulsion group. Two American prototypes flew this idea in 1954. The Lockheed XFV-1 never completed the cycle — its "highly tapered, straight-wing design made the transition to vertical flight only at altitude, using a jury-rigged, landing-gear cradle for conventional takeoff and landings" [1,2]. The Convair XFY-1 did: it flew vertically in August 1954 and "six transitions to conventional flight were successfully completed" [2].

Why the programme stopped matters, because the usual account is wrong. Two NASA reviews of United States V/STOL development — one written largely from the author's own flight-test experience [1] — judge the configuration itself favourably: "good configuration arrangement for low- and high-speed compatibility", with a high-speed potential near 500 mph. What they judge poorly is the machinery and the cockpit around it: "poor mechanical control system features including low actuator response rate"; "difficult to hover precisely over a spot"; "tip-over tendencies noted when on ground in gusty air" [1]. The landing difficulty is attributed to "the unusual spatial orientation where the pilot looked over his shoulder and down", to "the sensitivity to atmospheric turbulence", and to "reduced control power near touchdown" [2].

And then the reason testing ended, stated the same way in both:

> *"Six transitions to conventional flight were successfully completed* ***before testing was curtailed because of engine and gear-box reliability problems****." [2]*

The XFY-1 was not stopped by pilot workload. The workload was real, separately documented and severe — but what curtailed the testing was mechanical. **Three of the four objections recorded against these aircraft are objections to 1954 machinery and to a human pilot, not to the configuration**: actuator response rate, gearbox reliability, and an orientation problem that exists only because someone is sitting in it. Section 2.5 returns to what that leaves.

2.3 Distributing sweep along the span

The proposed planform varies its leading-edge sweep continuously from root to tip while holding the trailing edge at a constant angle. The principle of treating sweep, chord and thickness as one coupled distribution rather than three independent choices is not new; the crescent wing of the Handley Page Victor is its best-known expression, its sweep decreasing outboard so that the wing was not governed by its most vulnerable station.

The present aircraft is subsonic and does not inherit the transonic motivation that produced that planform. What it takes is the structural idea alone, and it takes it in a much reduced form: as Section 4.2 records, the sweep variation actually realised here is under seven degrees. No claim of descent from the crescent wing is made, and none is needed. Section 4.2 states the values used, and Section 8 states plainly that they were chosen rather than derived.

2.4 The contemporary hybrids

The problem did not go away when the prototypes did. Since roughly 2010 a large family of hybrid VTOL uncrewed aircraft has reached service, in two dominant architectures. Lift-plus-cruise aircraft carry a set of rotors for the vertical phase and a separate propulsor for cruise, and fly as a fixed-wing aircraft in between. Tilting architectures — tilt-rotor, tilt-wing and tilt-nacelle — reuse the same propulsors in both regimes by rotating them.

Both work. Both are in operational use. This paper does not claim otherwise, and Section 6.5 places the proposed reference designs alongside them without ranking them. What Section 3 argues is that each of these architectures pays for its capability in a way the other does not, that the payment can be moved between mass, drag and power system sizing, and that it cannot be brought to zero as long as hover and cruise are served by different hardware or by hardware that must move between two roles.

2.5 What this history does and does not show

It would be too convenient to declare that every one of these programmes ended for reasons external to its configuration. Some of the difficulties were real and internal, and this paper inherits them. The tail-sitter's vertical descent is genuinely harder than a runway landing. A tail-sitting aircraft is more exposed to crosswind on the ground than a conventional one. And a set of propellers whose thrust vectors are all parallel to the body axis cannot, by construction, produce a rolling moment — a limitation that applies to this configuration exactly as it applied to its predecessors, and which Section 4.4 addresses rather than avoids.

What the history does show is that the concept was never given a fair verdict under present-day conditions. The programmes of the 1950s were closed by engine deliveries, by jet-era procurement priorities, and — in the case where two contemporary NASA reviews record the reason directly and identically — by engine and gearbox reliability [1,2]. Not one of them was closed because the configuration failed to fly; one of the reviews rates the configuration itself favourably while condemning the machinery around it [1]. The human pilot, whose difficulties are the best-documented part of the record, is the single constraint that an uncrewed aircraft removes entirely. What is available now that was not available then — electric drive on each rotor, sensor-based attitude reference, and enough onboard computation that stability need not come from the airframe alone — removes exactly the obstacles that stopped it.

What the history does not settle is why the contemporary aircraft that did succeed still pay for their vertical capability, and what exactly they pay. That is the subject of the next section.

3. The architectural tax of hybrid VTOL

Hybrid VTOL aircraft work. The argument of this section is not that they do not, but that they pay for the capability in a way that can be located precisely, that the payment appears in three different currencies, and that an improvement in one currency is usually a transfer into another rather than a reduction. The section closes by stating the condition under which the payment would be zero — a condition none of the current architectures satisfies, and which Section 4 is built to satisfy.

3.1 The root: a duty cycle that does not match the hardware

Every hybrid VTOL aircraft contains hardware whose only purpose is the vertical phase. That phase is short. For a mission of one hour, a take-off, a transition, a return transition and a landing occupy on the order of a minute — roughly two percent of the flight. The remaining ninety-eight percent is spent carrying that hardware through the air.

This is not an implementation defect and it cannot be engineered away by making the hardware better, because it is a statement about duty cycle rather than about quality. A lighter lift rotor is still carried for the whole flight. A cleaner lift rotor is still carried for the whole flight. The mismatch between how long a component is needed and how long it is present is the origin of all three bills below.

3.2 Bill 1 — mass

The most direct payment is dead mass. A lift-plus-cruise aircraft carries two propulsion groups: rotors, motors, mounts, wiring and structural reinforcement for the vertical phase, and a separate propulsor for cruise. The vertical group is inert for the whole cruise but is still lifted. Its cost is not linear, because mass growth feeds itself: MTOW = m_payload / (1 − f_empty − f_energy) shows that additional empty mass enters through a multiplier that grows as the denominator shrinks, and in the vertical phase the same increment is charged again because hover power scales with W^1.5. A modest dead-mass fraction becomes a large payload penalty.

This bill has been identified independently. A NASA study sizing four VTOL architectures against a common mission with common tools found the lift-plus-cruise concepts the heaviest of the vehicles examined, and is explicit about the cause [22]:

> *"The weight of the Lift+Cruise concepts is heavier in general than for the other vehicles. This is not driven by the cruise power draw, as the L/D_e of the Lift+Cruise is indeed higher than the other vehicles... the most likely targets for reducing vehicle weight are the extra empty weight items on board in hover (wing and propeller)."*

The finding separates the two things this paper is at pains to separate: the lift-plus-cruise vehicle is *aerodynamically better* than the alternatives — its cruise efficiency is higher, and the study says so — and it is nevertheless the heaviest, because of hardware carried in order to hover. That is Bill 1 stated by an independent source in its own terms: not a failure of engineering, but the cost of an architecture.

A second NASA review states the structural half as a general principle, drawn from a tilt-prop aircraft whose propeller separated in flight after a gearbox mounting fatigued: "to safely transmit power to the extremities of the planform, very strong (and fatigue-resistant) structures must be incorporated with an obvious weight penalty" [2]. **Distributing lift or thrust across the span therefore obliges the structure that reaches it to be strong enough and fatigue-resistant enough to keep transmitting power there — charged to mass, whether or not the distributed propulsors are running.**

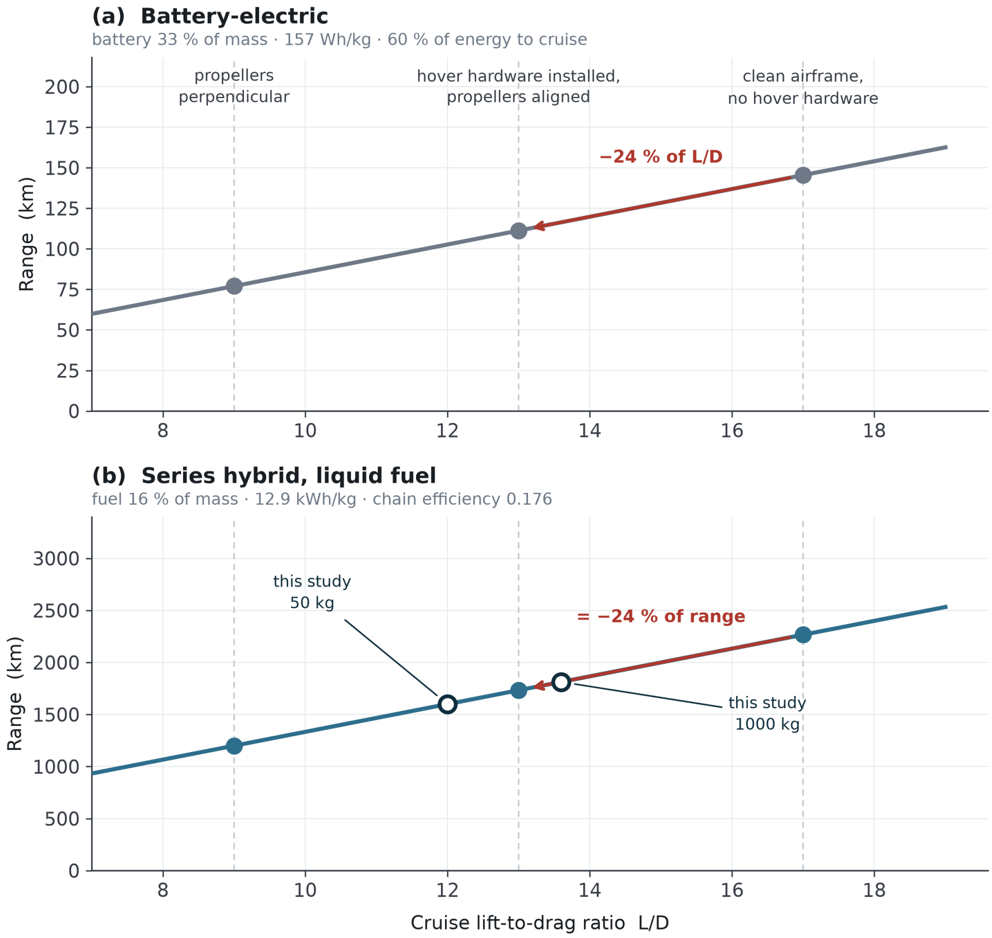
3.3 Bill 2 — drag

The second payment is charged only to architectures that leave hover hardware exposed in forward flight: rotors stopped in the airstream, the booms that carry them, and the interference between their wakes and the wing.

The cleanest measurement is a controlled comparison within a single aircraft. In a doctoral study, one uncrewed airframe was tested in a wind tunnel in four configurations:

Configuration	Maximum L/D
Clean airframe, no hover hardware	≈ 17
Hover hardware installed, propellers aligned with the flow	≈ 13
Hover hardware installed, propellers perpendicular to the flow	≈ 9

Two measured numbers follow. Installing the hover hardware costs about a quarter of the lift-to-drag ratio; failing to let the propellers align with the flow costs about a third of what remains. A second model on a different airframe reproduced the pattern, at L/D ≈ 11 retracted against ≈ 8 exposed. Expressed as drag, retracting the propellers reduced it by 34 % and 30 % relative to a standard quadplane. That study's author is careful about which comparison is legitimate — measuring the retracted aircraft against *itself* with propellers deployed gives 63 %, which he explicitly rejects — and the caution is worth adopting. Because range is linear in lift-to-drag ratio for a fixed energy system, this ladder translates directly into range (Figure 12).



Filled markers — L/D measured in wind-tunnel tests of one airframe in three of the four configurations tested.
Open markers — L/D calculated for the reference designs of this study; not measured.
Each line gives range for one fixed energy system: the markers locate L/D values only, and are not claims about the range of the aircraft those measurements came from.

Figure 12. Range against cruise lift-to-drag ratio for (a) a battery-electric and (b) a series-hybrid energy system. Filled markers are L/D values measured in wind-tunnel tests of one airframe [3]; open markers are the calculated reference designs of this study.

A third finding constrains what can be done about the penalty, and matters more than either number:

> *"The difference between propellers parallel to the airflow and without propellers is modest. The drag produced by the motors is significant."*

The bill is charged mainly by the motors and the beams that carry them — hardware that cannot be feathered, folded or aligned away, **because its cost is its presence.**

Two further measurements support the direction. Wind-tunnel characterisation of a QuadPlane found the highest lift and least drag in fixed-wing mode at both cruise airspeeds, drag in the hybrid regime exceeding either pure mode through adverse flow interaction, and — a point that matters for how such aircraft are designed — that a simulation assuming negligible rotor–structure interaction "always predicts higher lift and lower drag than were experimentally observed." Separately, a study of twenty-six stationary lift propellers held edge-on found their drag scaling with frontal area and the square of airspeed, with hover powertrain components adding "a significant amount of aerodynamic drag during forward flight" absent a stowing mechanism.

The important property of this bill is not its size but where it is charged. It is charged per unit time in cruise, so it grows with exactly the quantity the aircraft exists to maximise.

3.4 Bill 3 — power system sizing

The third payment is the least visible and often the largest. A VTOL aircraft must install enough power to hover, but it only uses that much power for the two percent of the flight in which it hovers. The ratio between the two demands follows from the governing equations rather than from any design choice. Taking hover power from momentum theory and cruise power from the drag polar,

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P_hover / W  = √(DL / 2ρ) / η_h                 (DL = W/A, disc loading)
P_cruise / W = V / ( (L/D) η_p )
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so that

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P_hover / P_cruise = √(DL / 2ρ) · (L/D) / V · (η_p / η_h)
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Every term on the right is a property of the configuration, not of the workmanship. A vehicle with a disc loading of 100 N m^{-2}, a cruise lift-to-drag ratio of 15 and a cruise speed of 30 m s^{-1} needs roughly four times as much power to hover as to cruise; raising the disc loading raises the ratio as its square root. The prediction is borne out in flight: a carbon-fibre tail-sitter reported in the literature measures its level-flight power consumption at one fifth of its hover power, which is the same ratio this expression gives for an aircraft of that class. The power system is therefore sized by a condition that holds for a minute and is then carried, unused, for an hour.

The consequence propagates. Sizing by hover means an oversized engine, or a battery that must deliver a peak it will rarely be asked for, or both — and whichever of the two is chosen, the extra installed capacity is mass, which returns to Bill 1.

3.5 The bills are one quantity in three currencies

The three bills are not independent problems with independent fixes. **Each known architectural move reduces one and increases another.** Figure 3 shows the transfers; Table 1 lists them.

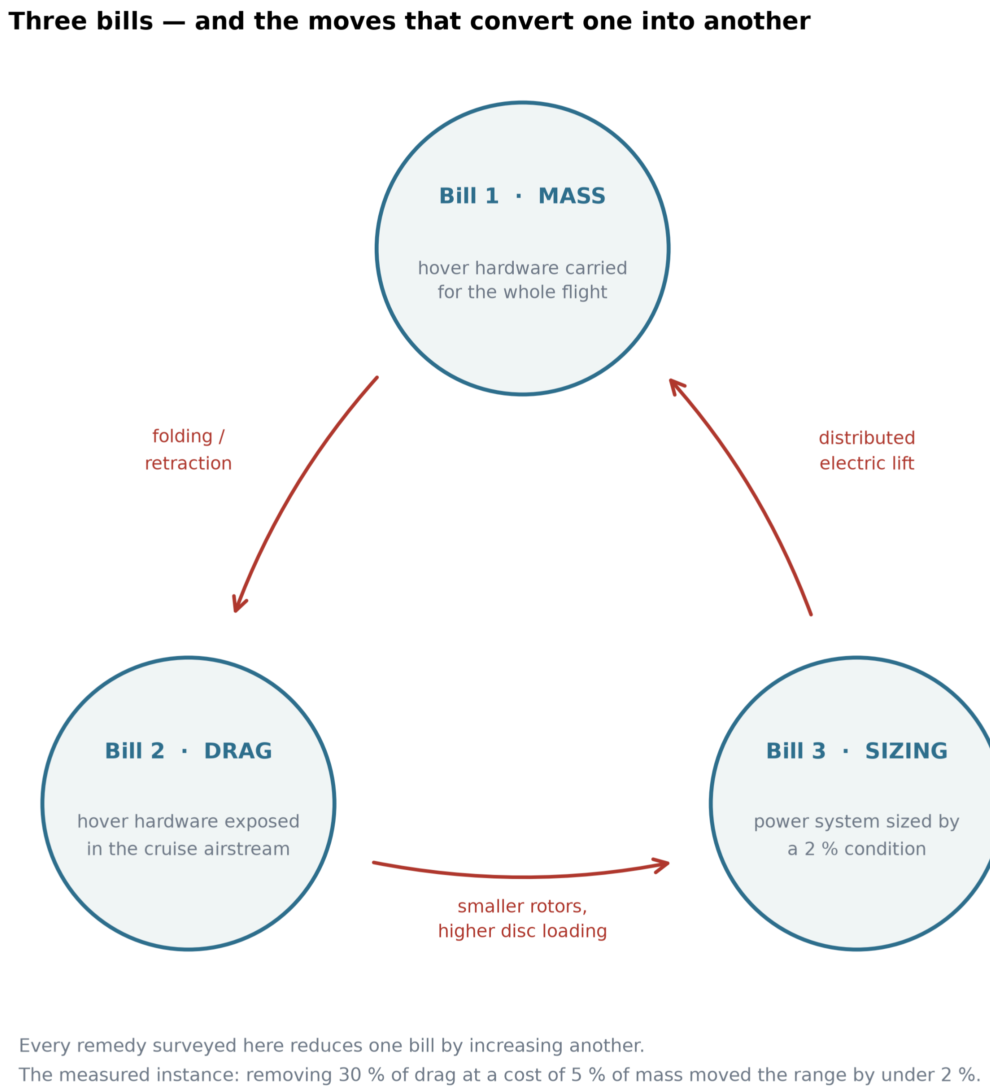

Figure 3. The three bills of the architectural tax and the moves that convert one into another. Every architectural remedy surveyed in Table 1 reduces one bill by increasing another. The measured instance quoted at the foot of the figure is derived in Section 3.5 from [3].

Table 1. Architectural moves and the bills they transfer.

Move	Bill it attacks	Bill it creates
Distributed electric lift rotors	3 — the cruise engine no longer sizes to hover	1 and 2 — many rotors and mounts, permanently carried and exposed
Folding or retracting lift rotors	2 — the exposed rotor is removed from cruise	1 — mechanism, actuation, locking, a new failure mode
Tilt-rotor, tilt-wing, tilt-nacelle	1 — one propulsion group serves both regimes	mechanical complexity, gyroscopic coupling, a transition control problem
Higher disc loading, smaller rotors	1 and 2 — smaller, lighter, cleaner rotors	3 — hover power rises with √(DL)
Lower disc loading, larger rotors	3 — hover power falls	1 and 2 — larger structure and exposed area

One of these rows has been measured. In the study of Section 3.3, the retraction system that removed thirty percent of the drag was then costed: applied to a passenger eVTOL with the mechanism assessed at five percent of vehicle mass, maximum range rose from 119 km to 121 km — **a two-kilometre gain for a five-percent mass penalty.** Bill 2 was converted almost exactly into Bill 1, and the transfer is the point rather than the small residue.

3.6 The condition under which the three bills are not charged

Stating the tax this way makes its escape condition explicit. The bill exists because hover and cruise are served by hardware that is not the same hardware, doing the same job, in the same orientation. Relax any part of that and a bill appears:

- **Different hardware** → Bills 1 and 2. The unused set is carried and, if exposed, drags.
- **Same hardware, different orientation** → the tilting family. The mechanism that changes the orientation is itself mass, complexity and a control problem.
- **Same hardware, same orientation, different sizing point** → Bill 3, unless the hover peak is supplied from somewhere other than the continuous power source.

The condition for not incurring these three bills is therefore that the same propulsors, fixed in the same orientation relative to the airframe, produce both the hover thrust and the cruise thrust — with the aircraft itself changing orientation rather than any part of it — and that the difference between the hover peak and the cruise demand is supplied by a buffer rather than by permanently installed continuous power. This is referred to below as the *zero-bill condition*, and the name should be read strictly: it means zero of these three bills, not an architecture that costs nothing. What an architecture satisfying it pays instead is a separate question, and Section 5.4 answers it for the configuration proposed here.

That is a description of a tail-sitter with a buffered series-hybrid powertrain. It is also, precisely, the configuration described in Section 4.

3.7 The three bills stated formally

Supplementary S6 states the three bills as equations and gives the transfer table that shows them to be one quantity in three currencies: a design that refuses to pay one of them pays it in another. The short form is that each bill is a fraction of take-off mass, exposed cruise drag, or installed continuous power, and that the three are linked by the sizing loop — mass drives thrust, thrust drives power, power drives mass — so that relieving one without relieving its cause simply moves the charge.

A consequence that can be checked against published work. If the framework is right, then for the same mission a configuration carrying a dedicated lift system should pay for it in gross weight, and that payment should *not* be recovered by the cruise efficiency the arrangement buys. This is a sharper prediction than it first appears, because it forbids the obvious defence: it says the efficiency gain is real and still insufficient. The NASA sizing set is a direct test of it. Against a common mission of 1 200 lb of payload over

75 nautical miles, the turboshaft quadrotor — which has no cruise wing, and therefore no dedicated lift hardware to carry — sizes at an effective lift-to-drag ratio of 4.9 and a design gross weight of 3 678 lb, while the turbo-electric lift-plus-cruise reaches 8.5 and weighs 7 271 lb [22]. **Its cruise efficiency is seventy percent better and it is nearly twice as heavy**, which is the prediction and not a counter-example to it. The tilt-wing in the same set reaches 8.6 — higher than every lift-plus-cruise entry — while carrying no dedicated lift system at all, and it is the one configuration in the table that uses the same hardware in both regimes. The framework does not predict the numbers; it predicts that the weight charge survives the efficiency credit, and in this set it does.

What the framework does not claim. It does not claim that avoiding the three bills makes an aircraft better, only cheaper in those three specific currencies. A configuration may avoid all three and still be unbuildable, uncontrollable, or unsuited to its mission — and Sections 7 and 8 are about exactly that possibility for the configuration proposed here. Nor does it claim the bills are the only costs; they are the ones that follow from the duty-cycle mismatch of Section 3.1, and a design pays many others.

3.8 Why the market looks the way it does

One observable consequence supports the argument, and it has been stated independently. Surveying the field, the study cited above concludes that multirotors are efficient in hover and suited to short-range missions, that vectored-thrust aircraft are efficient in cruise and suited to long-range missions, and that "lift plus cruise eVTOLs are a compromise, but they are slowed down by the drag of the lift propellers."

Hybrid VTOL aircraft occupy a narrow band of the mission space. Below it, where range requirements are short, a multirotor is cheaper and simpler and pays none of these bills because it never claimed cruise efficiency. Above it, where range requirements are long, a runway-launched fixed-wing aircraft is more efficient and pays none of them because it never claimed vertical capability. The hybrid sits between the two, and the width of that band is set by how much cruise efficiency the architecture had to surrender.

An architecture that surrenders less does not merely perform better inside the band. It widens the band.

4. Proposed configuration

4.1 Overview

The configuration satisfies the zero-bill condition of Section 3.6 directly rather than by compensation. The aircraft stands on its tail. Its entire airframe is a blended-wing body: there is no cylindrical fuselage, and every part of the planform carries payload and produces lift. A single coaxial counter-rotating propeller pair at the nose produces all propulsive thrust, in hover and in cruise alike, without changing its orientation relative to the airframe. Four small coaxial pairs, mounted on rigid frames at the wing tips, produce attitude control and nothing else. The energy system is a series hybrid: fuel drives an internal-combustion engine, the engine drives a generator, and the generator supplies electric machines at the rotors.

The aircraft has no elevons, no rudder, no tilting mechanism, no retraction mechanism and no dedicated lift system. The only moving aerodynamic device is a variable-extension strip on the lower surface, described in Section 4.4, which exists solely because roll cannot be produced by propellers alone. Figure 4 gives three orthogonal views of the light reference design and Figure 5 a general view of the same geometry.

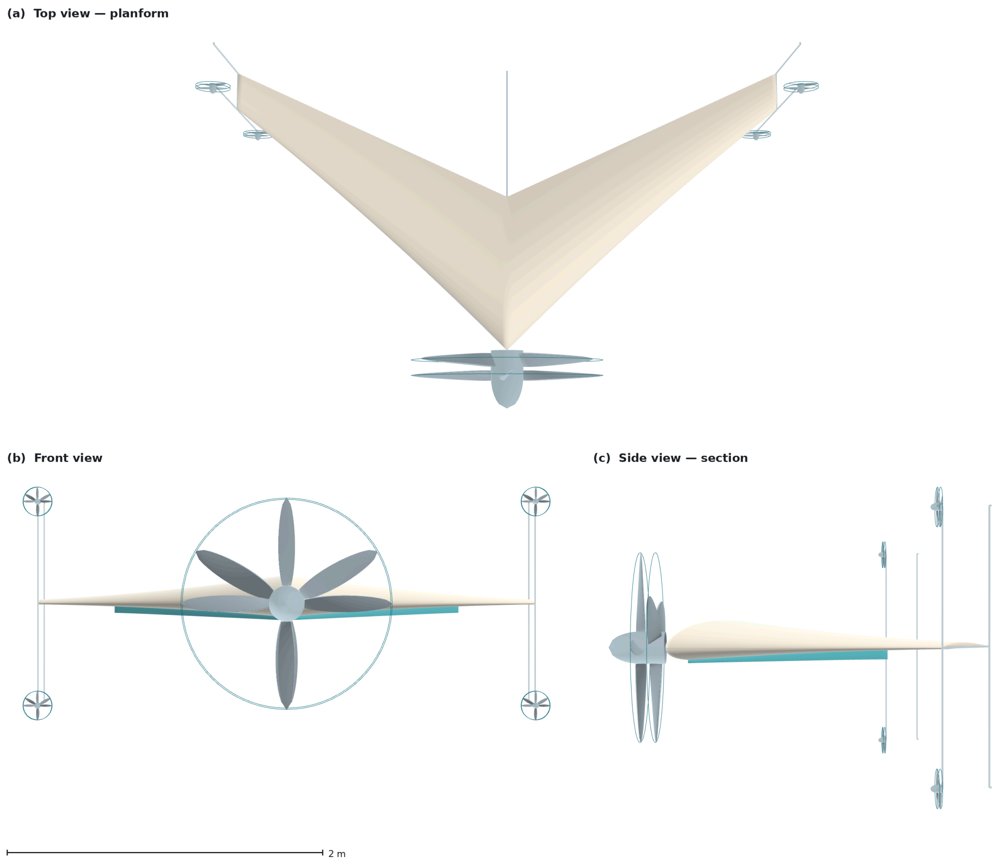

Figure 4. The light reference design, 50 kg, in three orthogonal views: (a) planform, (b) front view, (c) side view showing the aerofoil section and the tip frames. Rendered from the parametric geometry model; the scale bar applies to all three views.

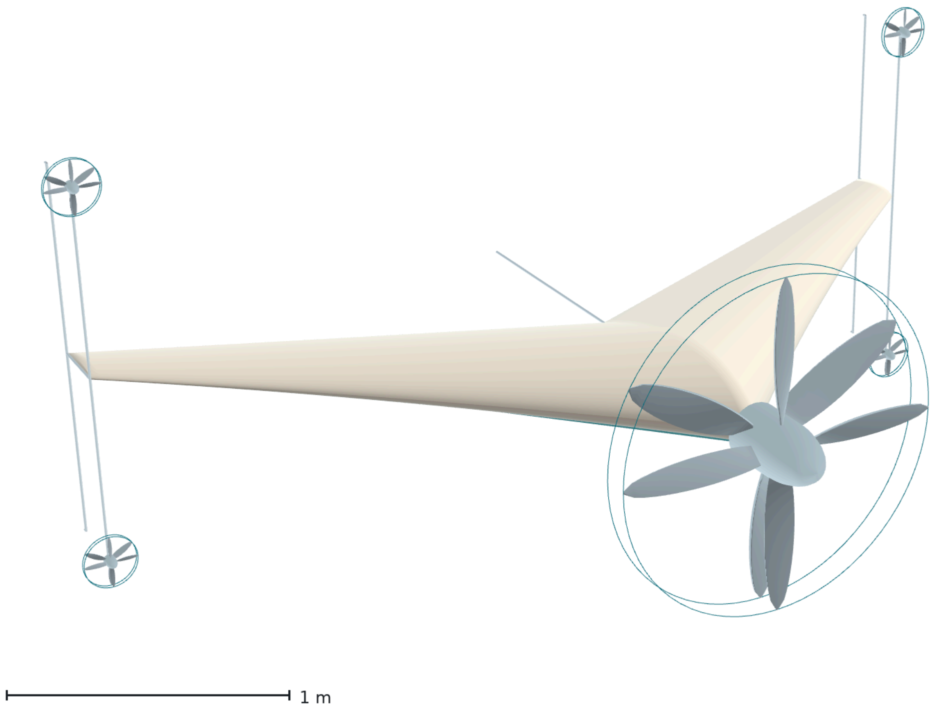

Figure 5. General view of the same geometry. The single coaxial pair at the nose, the four small coaxial pairs at the ends of the tip frames, and the absence of any control surface are all visible.

4.2 Planform

The planform is a blended-wing body whose leading-edge sweep varies continuously along the span while the trailing edge is held constant. The sweep law is linear in span from 45° at the root towards 35° at the station where leading and trailing edges would converge; the wing is cropped at 67 % of that station, so the realised sweep runs from **45° at the root to 38.3° at the tip**, with the trailing edge constant at **25°**. Thickness runs from 25 % of chord at the root to 12 % at the tip and the chord from 0.970 m to 0.236 m, a taper ratio of 0.244 — one distribution rather than four independent ones. **The realised sweep variation is therefore modest, under seven degrees**, and the crescent character comes from the curvature of the leading edge and the divergence between the two edge angles rather than from a large change in sweep. These values were chosen, not derived: the 20°–40° band they started from was reported favourable in the transonic-transport literature the crescent-wing idea comes from, and this aircraft is subsonic, so that band does not transfer on its own authority. Section 8 lists it among the limitations.

Sweep does two jobs here, and the second is why it is not a free parameter. The aircraft is tailless: with no horizontal stabiliser on a boom, the pitching moment must be generated by the distribution of lift along the body itself, and sweep is what places the outboard sections behind the centre of gravity so that they can do it. **The sweep angle and the longitudinal stability are the same design variable seen from two directions.** Decreasing sweep outboard also keeps the tips from stalling first, which matters more than usual on a tailless aircraft because a tip stall on a swept planform moves the centre of pressure forward and pitches the aircraft further in, with no tail to argue with.

The reference geometry for the light design is a root chord of 0.970 m, a tip chord of 0.236 m, a span of 3.453 m, a wing area of 1.979 m² and an aspect ratio of 6.03, giving a wing loading of 25.3 kg m⁻² and a stall speed of 20.1 m s⁻¹ against a cruise speed of 30 m s⁻¹. Figure 6 gives the distributions.

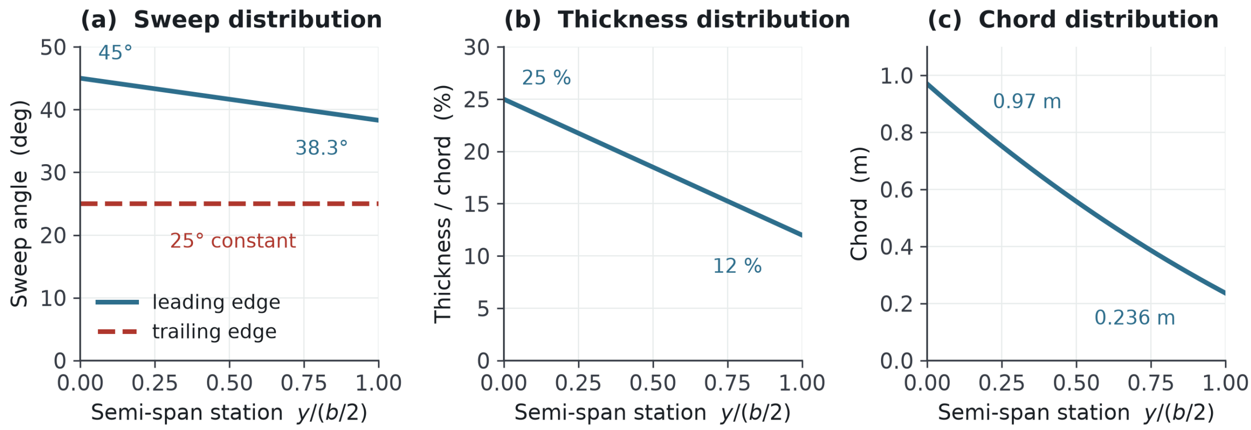

Figure 6. Spanwise distributions of the planform. (a) Leading-edge sweep falls from 45° at the root to 38.3° at the crop station while the trailing edge is held constant at 25°; (b) thickness-to-chord ratio; (c) chord. These are one coupled distribution rather than three independent choices.

4.3 Propulsion

Every propeller is a coaxial counter-rotating pair, for one reason worth stating narrowly: **reaction torque.** A single propeller applies to the airframe a torque equal and opposite to the one it applies to the air, acting about the yaw axis in hover and the roll axis in cruise, and it must be opposed continuously — by a control surface, which costs drag, or by differential thrust, which costs a control channel. A counter-rotating pair does not produce it.

The pairs are of fixed geometry: no cyclic pitch, no collective, no variable mechanism. **This has a consequence for the tip pairs in cruise that Section 5.2 works out** — unable to feather, they must either turn at the zero-shaft-torque condition or be stopped, and the difference between those two states is most of the aircraft's zero-lift drag. The torque balance is exact at cruise rather than hover, so a small residual remains in hover.

The arrangement has a second consequence the transition analysis depends on. Because the two rotors of each pair carry equal and opposite angular momentum, **the net angular momentum of the propulsion system is nominally zero**: rotating the airframe through ninety degrees precesses nothing, and no gyroscopic moment appears for the control system to cancel. In a tilting architecture that term is present and must be designed for; here it is absent by construction.

Crucially, the pair is never a contra-rotating gearbox. Each rotor is driven by its own electric machine on a common axis, so the mechanism that repeatedly defeated the XB-35

— concentric shafts, a splitting gearbox, and the governors that synchronise them — is never built.

The energy path is a series hybrid: fuel → engine → generator → electric machines. The engine is not mechanically connected to any rotor; it is an energy source, and that decoupling is what allows it to be sized for cruise rather than hover. For the light design the continuous cruise requirement is 1.9 kW at the engine shaft and the engine is sized at 2.6 kW, while the hover requirement is 10.9 kW at the rotor; the difference is supplied for the vertical phase by a **1.8 kg battery buffer, 3.6 percent of take-off mass**. That is the mechanism that releases the engine from the hover condition — Bill 3 as Section 3 defines it. The electrical path is not released, and Section 5.3 says so.

4.4 Control without control surfaces

The nose pair produces thrust; the four tip pairs produce moments. They are not lift rotors and are not sized to hover the aircraft — in the vertical phase they carry under fifteen percent of total power. Each tip pair on the light design is 0.20 m in diameter and produces 16.2 N during the transition manoeuvre, drawing 335 W, 1.34 kW across the four. The principle is not new: a survey of tail-sitter development identifies adding propellers away from the axis as a way to generate transition pitching moment, and a flight-tested quad tail-sitter uses exactly this arrangement [30]. The full derivations for all three axes are in Supplementary S3; what follows is the result in each.

Pitch and yaw are produced by differential thrust, and the yaw arm is the larger. The tip frames extend ±0.71 m perpendicular to the planform, so a differential between the upper and lower pairs acts at 0.71 m in pitch, while a differential between the left and right pairs acts at the **semi-span, 1.726 m — 2.43 times the pitch arm**. Yaw is therefore the strongest axis on this aircraft, which is the reverse of the usual situation and is a free consequence of the tip-propeller layout rather than a design choice. Figure 7 shows the placement and the two arms, and shows why the third axis has neither: every thrust vector is parallel to the body axis, so no combination of settings produces a rolling moment.

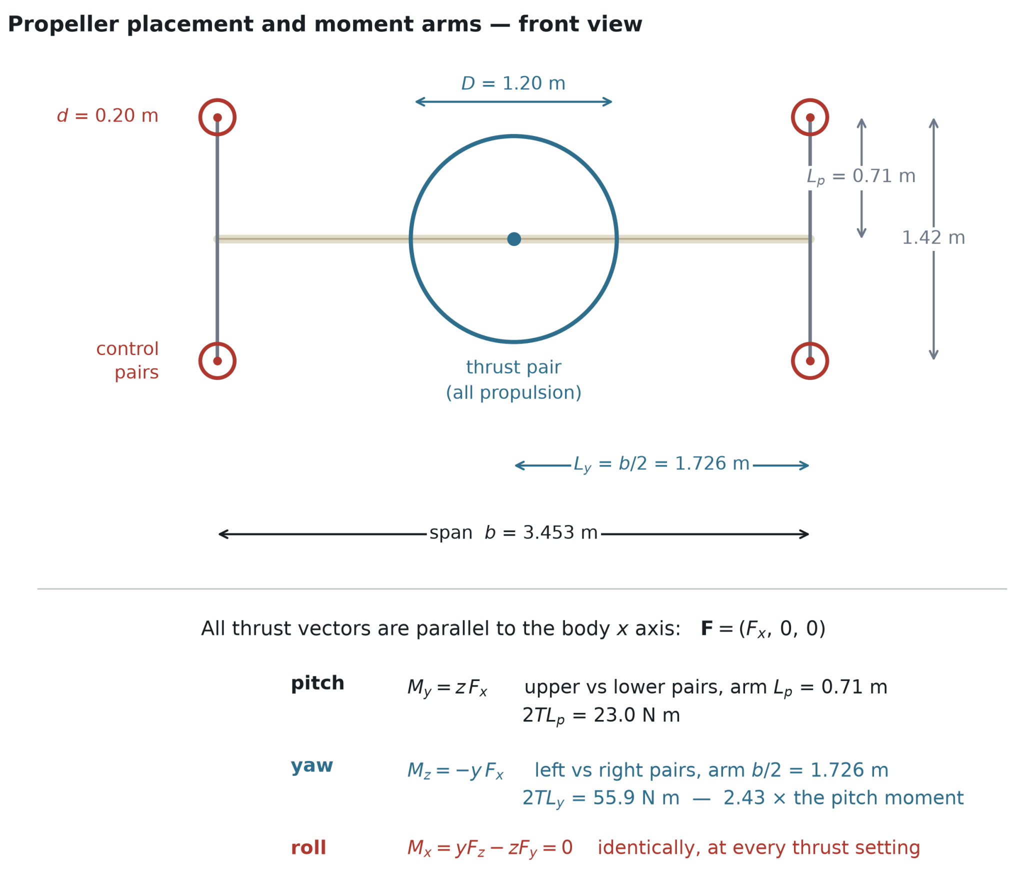

Figure 7. Propeller placement and moment arms, front view. Every thrust vector is parallel to the body x axis. Pitch and yaw follow directly from differential thrust; the rolling moment is identically zero at every thrust setting, which is the gap the strip of Figure 8 exists to close.

Roll cannot be produced by propellers at all, because every pair is coaxial and torque-balanced by construction. It is the one axis that needs an aerodynamic device, and that device is the only moving aerodynamic surface on the aircraft: a strip on the lower surface, inclined at 45° in planform, running 120 % of root chord and reaching 67 % of semi-span, standing 2 cm proud at its inboard end and 6 cm at its outboard end. **Extension is the control variable** — the strip is modulated, not switched. Figure 8 shows it against the nose propeller's slipstream: the inboard 46 percent of its length lies inside the slipstream, where dynamic pressure is set by disc loading and is therefore available at zero airspeed, and the outboard 54 percent works against the freestream in cruise. That split is why one device serves both regimes.

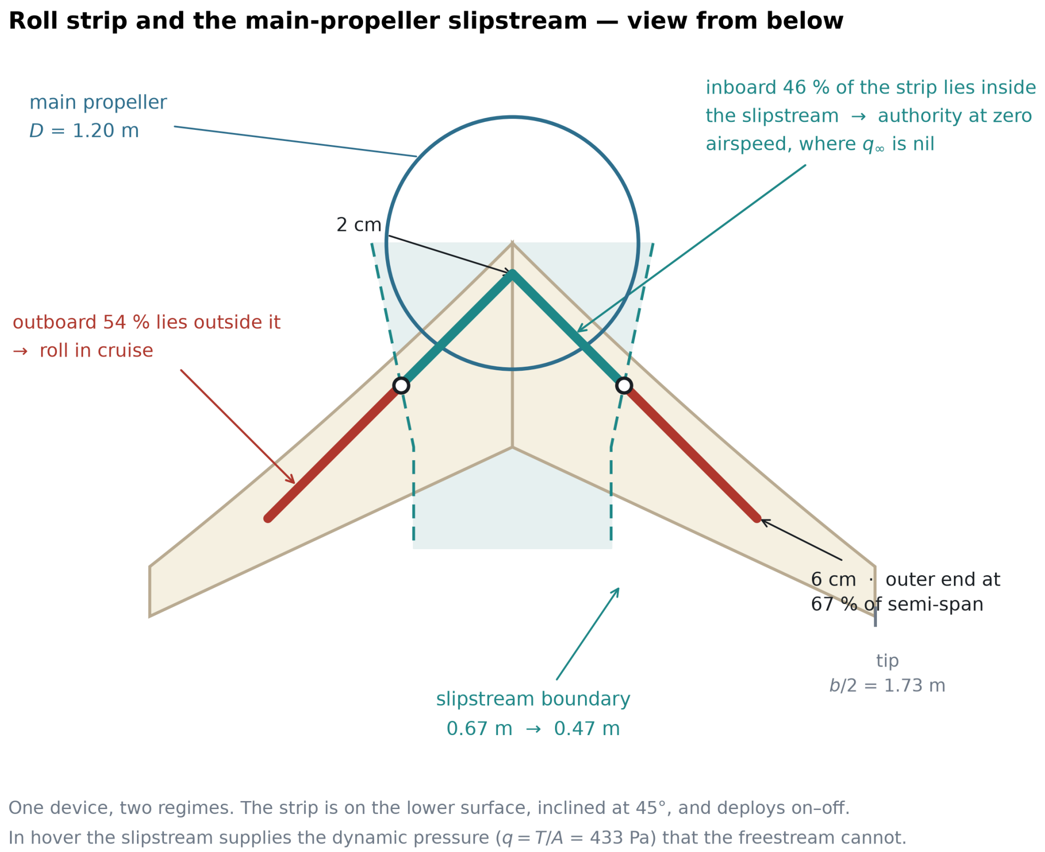

Figure 8. The roll strip against the main-propeller slipstream, viewed from below. The inboard 46 % of the strip lies inside the slipstream, where the dynamic pressure is set by disc loading and is therefore available at zero airspeed; the outboard 54 % lies outside it and works against the freestream in cruise.

Roll inertia computed from the component mass distribution is 25.0 kg·m^2, two and a half times the pitch inertia, and the roll damping derivative from a helix-angle vortex-lattice solution is $|C_{l_p}|$ = 0.358, giving a damping slope of 77.6 N·m per rad s^{-1} and a roll time constant of 0.32 s. **Twenty degrees per second at cruise therefore requires 27.1 N·m.** The strip's own force as a swept fence, at an upper-bound normal-force coefficient of 1.3, supplies 11.3 N·m — about a third. The remainder must come from the second mechanism, the change the strip makes to the circulation of the half-wing it sits on, which asks for **$\Delta C_L \approx 0.12$** over the strip's span. Measured data on non-planar wings carrying Gurney flaps of two percent chord over the inboard two-thirds gives lift increments of that magnitude [18], and a lift-enhancing tab study gives a measured height threshold of 1.5 % chord above which maximum lift-to-drag falls [32]. **Roll authority is therefore sized and supported by measurement on a comparable device, and it is not closed**: the quantity a future measurement must return is ΔC_L for this strip on this planform.

The strip loads the other two axes, and Supplementary S3 quantifies all three couplings. Deployed on one side it raises drag there and yaws the aircraft towards the strip — adverse yaw in the classical sense, 11.3 N·m at the present height law, against 42.8 to 55.9 N·m of yaw authority, so it costs five to twenty-six percent of the strongest axis. It also pitches the nose down, by ΔC_m between 0.005 and 0.032 depending on where along the chord the lift increment acts; the tightest pitching-moment budget in Section 7.6 is 0.050, so at the upper end a commanded roll would consume two thirds of it.

The conclusion is a scheduling requirement rather than a redesign: the strip should not be commanded at full extension during the end of the rotation. The older literature states the same coupling more bluntly — for spoilers used as ailerons on tailless aircraft, "if only upgoing spoiler projections are used, the pitching moments developed are prohibitive", with projection from both surfaces named as the remedy [19]. The strip here projects from one surface only, which is the arrangement that warning is about.

Because extension is continuous, the strip also has a threshold. Projections below one percent of local chord produce negligible lift change [19], and the strip is tapered, so this cuts a graded dead band out of the bottom of its travel: **nothing below seven percent of commanded extension, and linear to within five percent only above twenty-five**. The part that survives the threshold is the part with the longest arm, so most of the lost area is bought back by the lengthened arm — the deficit is confined to small corrections rather than large ones.

The same device has a third use that costs nothing. Deployed on both halves at once it is a speed brake, and in that mode the pitching-moment objection does not arise: spoiler-type ailerons used as speed brakes "had only a small effect on the wing pitching moments", and "the rolling effectiveness of the ailerons will not be impaired by such use" [49]. That gives this configuration a descent-rate control no other part of it provides.

Hover is the harder case, for a structural reason rather than a numerical one. At zero airspeed only the inboard part of the strip is loaded, by the nose propeller's slipstream, and the same circulation model over the slipstream-washed area gives 6.0 to 12.0 N·m — enough for a thirty-degree bank in 1.5 to 2.1 s. But there is no aerodynamic damping at all at zero airspeed: the roll axis is a double integrator, so the rate does not settle and the strip must be commanded off rather than left out. **Roll control in hover is a tighter problem than roll control in cruise**, which is the reverse of the usual situation.

Two measured effects work against the strip in the conditions this configuration needs it, and one works for it. A Gurney flap under controlled inflow turbulence became "less effective after stall angle", and at nineteen percent turbulence intensity "the benefit to the aerodynamic performance was negligible" [35] — and the strip's inboard portion is deliberately placed inside a propeller slipstream, which is not a low-turbulence environment. The post-stall half of that is contested: a delayed-detached-eddy simulation at twenty degrees of incidence with a flap of the same height fraction returns lift higher by ninety-four percent at unchanged drag [37], though on a baseline twenty-five percent below the experimental value. **The turbulence sensitivity is measured and unopposed; the post-stall behaviour is contested.** Against both, a water-tunnel study found the mechanism qualitatively unchanged at a Reynolds number of 8 588 — four orders of magnitude below the usual test range — because an effective camber increase is "an inviscid effect to the first order" [38]. That is the most direct support available for the claim that the strip still works at zero airspeed, and it supports the direction rather than the magnitude.

Directional stability is the one open half of the yaw axis. A vortex-lattice solution of the planform at sideslip returns **$C_{n\beta} = 0$**: the wing supplies none, which is not a defect of the solution but the expected result for a planar surface, and which two measurement campaigns confirm is normal for tailless aircraft rather than unusual [19,20]. Sweep gives this configuration its roll-due-to-sideslip — $C_{l\beta} = -0.045$ per radian, a healthy value — and gives it no weathercock stability at all. **Zero is also an optimistic starting point rather than a neutral one**, because a vortex-lattice model has no volume: the centre body develops side force ahead of the centre of gravity, and on tailless aircraft that destabilising contribution is reported to be "at least as great as the stabilising effects contributed by the wing alone" [19].

Directional stability must therefore come from the tip frames, which in cruise are the only vertical surfaces the aircraft has. Sized against the criterion the tailless literature recommends — $C_{n\beta}$ greater than 0.001 per degree, or 0.0573 per radian [19] — the fairing chord required over the combined frame length is **39 mm**, against the 50 to 70 mm a 20 mm faired strut carries in any case. **Directional stability on this configuration does not ask for a surface; it asks for a fairing on a frame that is already there.** Two requirements come with it. The fairing needs a toe angle, and its sign follows from the aspect ratio: low-aspect-ratio fins are toed *in* so the stabilising moment comes from induced drag, high-aspect-ratio fins toed *out* so it comes from "the outwardly directed lift" [19] — and these frames are firmly in the second class, so **the sign is toe-out**. Toe-out carries a hazard: yawing far enough to stall the rear fin produces a large *destabilising* moment, which sets an upper bound on usable sideslip that this paper does not compute.

The second requirement is harder, and the measured record is against the assumption the fairing was sized on. A 39 mm chord at cruise sits at a chord Reynolds number near 80 000, and the side force credited to it above was computed from a lift-curve slope of 4 per radian. Symmetric sections at that Reynolds number are not measured to behave that way. The compilation that reports four such sections tested at Princeton finds lift-curve nonlinearity about zero incidence in all of them, in the severest case the slope "actually changed sign over a 3 deg range" [36] — and it states the effect as a property of the class rather than an accident of one section: "past work on **symmetrical** airfoils has shown that a **deadband often appears in the lift curve near zero degrees**", present at Reynolds numbers of 60 000 and 100 000 and absent at higher ones [36]. A toe angle of one or two degrees puts the fairing inside that band. **The open question is therefore not which way to toe the fairing but whether a surface of that chord, at that Reynolds number, develops the side force at all** — and 4 per radian should be read as an upper bound rather than as a conservative choice, since what fails there is the linearity of the curve and not merely the size of its slope.

The same source names the remedy, and it is cheap here. Camber removes the deadband: cambered sections "do not appear to have a similar, intrinsic deadband region" [36], and the observation is drawn from designing tail surfaces, which is what these fairings are. A fairing that is cambered outboard and toed out would take its side force from the linear part of a curve that has one, at no cost the frame does not already pay. This paper does not size that fairing, because doing so on a curve it has not measured would repeat the error it has just described. Supplementary S3 states it as the measurement this configuration would buy first.

4.5 Structure and ground contact

The tip frames are not added for the propellers. They are the landing structure.

This inverts a cost into a saving, and the inversion has precedent. Reviewing the tail-sitters of the 1950s, NASA noted that "dispensing with a conventional landing gear improved the empty weight fraction for these VATOL aircraft", while adding that some form of gear was still required on the tail surfaces [2]. The present configuration takes the same benefit and extends it: the structure that meets the ground is also the structure that carries the control propellers and sets their moment arm.

The aircraft rests on five points: the four lower ends of the tip frames and a single keel that runs aft along the centreline from the nose propeller to the trailing edge. Because the aircraft stands on its tail, these five points are what it stands on, and their spread is the stance base that resists tipping in wind. Lengthening the frames to buy the control moment arm of Section 4.4 widens the stance base at the same time. One structure serves three purposes — mounting the control propellers, providing the moment arm, and carrying the landing loads — and is charged to the mass budget once.

The stance base is a design parameter, not a constraint imposed by the configuration. Moving the frame ends further outboard widens it without altering the planform, the propulsion, or the control architecture — and because the same displacement lengthens the control moment arm of Section 4.4, the two benefits arrive together from one change. The reference geometry given here is one point on that trade; an operator with a stronger ground-wind requirement can take another without redesigning the aircraft.

5. The architectural tax, audited bill by bill

Section 3 identified three bills and argued that they are one quantity paid in three currencies. Section 4 described a configuration built to satisfy the zero-bill condition. This section audits that claim bill by bill, and then states, in the same detail, what the configuration does pay. The second half is not a concession appended for balance. An architecture that claimed to pay nothing would be describing a different aircraft from the one in Section 4.

5.1 Bill 1 — mass: the charge does not arise

There is no second propulsion group. The nose pair that lifts the aircraft off the ground is the same pair, in the same orientation, at the same station, that propels it in cruise. Nothing is carried unused.

This is the whole of the argument, and its brevity is the point. The bill was never a consequence of poor design in lift-plus-cruise aircraft; it was a consequence of counting two propulsion systems where the mission needs one. A configuration that counts one does not reduce the bill — it does not generate it.

The tip pairs are a genuine addition and are accounted for in Section 5.4. They are not a second propulsion group: they are sized for moments rather than for weight, and in the vertical phase they draw 1.34 kW against the nose pair's 10.9 kW, which is twelve percent.

5.2 Bill 2 — drag: reduced, not removed

In cruise there is no stopped rotor in the airstream, because there is no rotor that stops. The nose pair is the cruise propulsor and runs at its design condition throughout. The quarter of lift-to-drag ratio that Section 3.3 reports as the measured cost of installing hover hardware is not incurred — not reduced, not mitigated, but **absent**, because the hardware that causes it does not exist here. Nor is there a retraction mechanism, so the transfer of Bill 2 into Bill 1 does not occur either. Where lift rotors are retained, keeping their drag small needs an indexing mechanism to stop them at a favourable azimuth, or a retraction mechanism to stow them; both are mass and both are failure modes, and a configuration with no rotor to stop needs neither.

That carries a condition the paper had not stated, and it is a sharp one. The four tip pairs *are* rotors, and the claim holds only if they do not stop in cruise. Their eight discs sweep 0.251 m², **12.7 percent of the wing area** — not a small object to leave in the airstream in the wrong state:

Tip rotors in cruise	ΔC_D0	of the 0.0248 assumed
turning at zero shaft load, blades at low incidence	0.0003 – 0.0008	1 – 3 %
stopped edge-on, at a chosen azimuth	0.0008	3 %
stopped broadside, azimuth uncontrolled	**0.015 – 0.018**	**61 – 74 %**

The difference between the first and third rows is the difference between a configuration that works and one that does not, and it is robust to the coefficients assumed, because the two states differ by a factor of thirty. The second row shows why stopping them is survivable only if azimuth is controlled — which is the indexing mechanism this section has just claimed the configuration does not need.

The resolution costs nothing, and it is a control state rather than hardware. A fixed-pitch propeller left free settles at the advance ratio where net shaft torque is zero: inner sections drive, outer sections retard, and they balance. The shaft then does no work, so the motor neither drives nor brakes and the electrical cost is controller standby draw and bearing losses. The blades sit at low incidence with attached flow, which is the first row. **The tip rotors are therefore held in cruise at the zero-shaft-torque condition — neither stopped nor driven** — and this is the state assumed throughout Section 6. It is worth naming because both neighbouring states are wrong: driven, they cost propulsive power; stopped without azimuth control, they cost most of the aircraft's zero-lift drag.

What Bill 2 *is* paid, and this is why the heading says reduced rather than removed, is the tip frames. They are structure in the airstream that a conventional aircraft does not carry, and at the light design point they contribute ΔC_D0 = 0.0043 — **about twelve percent of total cruise drag**, and seventeen percent of the zero-lift drag the sizing assumes. Both denominators appear in this paper and each is named where it is used. That is the honest figure and it is carried in the ledger of Section 5.4.

The fairing on those frames is not only a drag measure. The frames are the only surfaces standing perpendicular to the wing plane, and the planform supplies no directional stability at all, so the fairing is also the vertical surface that provides it. Sized against the criterion the tailless literature recommends — C_n_β greater than 0.001 per degree [19] — the chord required over the combined frame length is 39 mm, against the 50 to 70 mm a 20 mm faired strut carries in any case. The two requirements do not conflict, and the directional one is the looser; but the frame cross-section is now constrained from two directions, and a selection made on drag alone would be made on half the evidence.

5.3 Bill 3 — power system sizing: avoided for the engine, not for the electrical path

The series-hybrid arrangement of Section 4.3 breaks the link that forces the power system to be sized by the hover condition. Because the engine drives a generator rather than a rotor, it supplies average power, not peak power, and the peak is supplied from a buffer.

For the light reference design the numbers are as follows. Cruise draws 1.7 kW at the electric machines, which is 1.9 kW at the engine shaft once the generator and power electronics are accounted for, and the engine is sized at 2.6 kW. Hover requires 10.9 kW at the rotor — 4.2 times the engine's rating. The difference is drawn for the duration of the vertical phase from a 1.8 kg battery, which is 3.6 percent of the maximum take-off mass. **That difference must be taken at one station, and it is the electrical bus**: the rotor's 10.9 kW of shaft power is 12.47 kW at the bus once the machine and the power electronics are passed, the engine delivers 2.34 kW there through the generator, and the buffer supplies the remaining 10.13 kW. Supplementary S2 gives the chain and records that an earlier version of this paper differenced two shaft stations instead, understating the demand on the buffer by twenty-two percent. The heavy reference design sits on the same line: 39.2 kW electrical in cruise, 54.3 kW engine, 216.2 kW hover, 40 kg of battery at 4.0 percent of MTOW.

An aircraft of this class whose powerplant had to be sized for hover would carry an engine rated above 10.9 kW instead of 2.6 kW. The mass difference is not recovered elsewhere; it is simply not incurred. That the buffer costs under four percent of MTOW at both design points, twenty times apart in mass, is the numerical statement that this avoidance is architectural rather than a fortunate coincidence of one size.

What is not avoided, and the section heading says so. The bill is defined in Section 3 as a *continuous* power system sized by the hover peak, and it is the engine and its fuel consumption that the buffer releases from that condition. The electrical path is not released: the nose motor and the power electronics must still pass the full 10.9 kW, and the component build-up of Section 6.7 shows them as 2.73 kg and 0.61 kg against 2.60 kg of engine and generator — that is, the hover-sized electrical machine is the single largest item in the propulsion chain. The saving is real and it is the engine's, but a reader should not take it as an aircraft on which nothing is sized by hover. The buffer itself carries a further condition, given in Section 6.7: it is specified by power rather than energy, at **5.63 kW kg⁻¹** to hover and 6.48 to leave the ground, which is a demanding cell requirement and not a free parameter. Section 8.2 measures it against what has been flown.

5.4 What is paid

The honest ledger has five entries.

The control propellers. Four pairs, their motors, mounts and wiring exist only to produce moments. In the vertical phase they draw twelve percent of the power the nose pair draws. This is the configuration's substitute for elevons and a rudder, and it is not free — it is merely cheaper than a second lift system, and it does not sit in the cruise airstream in the way a lift rotor does.

The tip frames. As established in Section 5.2, of the order of twelve percent of cruise drag, conditional on being faired. This is the largest single payment the configuration makes, and it is the price of the moment arm, the propeller mounting and the landing structure combined into one member.

The roll strip. The one moving aerodynamic device on the aircraft. Its cost when retracted is a surface discontinuity; when deployed it is a drag device by construction, but it is deployed only while a roll is being commanded.

The twist needed to trim. Section 7.6 finds that the configuration trims at cruise with nine degrees of tip washout, reflex being an order of magnitude short of the moment required. Washout is not free: it costs span efficiency, and the cruise lift-to-drag ratio falls from 12.65 to 12.11 — **4.3 percent**. This entry was missing from earlier versions of this ledger, and it is worth being precise about what it is a payment for. It is not one of the three bills of Section 3, which are charged for having a hover capability; it is charged for being tailless, and a tailed aircraft of the same architecture would not pay it. It belongs here because this configuration is tailless, and because a ledger that omitted it would be flattering rather than honest.

The transition manoeuvre. The aircraft must rotate through ninety degrees, and the rotation costs time, horizontal displacement and control power. Section 7 treats it in full and shows that the altitude cost, which is the one usually assumed to dominate, can be brought to zero: rotating slowly and entering the rotation while still climbing removes it entirely at both design points. What remains is not free — the manoeuvre occupies seconds during which the aircraft is neither hovering nor cruising — but it is smaller than the literature on tail-sitters would suggest, and it is the one payment on this list that gets *cheaper* the less it is hurried.

Set against the bills of Section 3, the ledger is favourable but not empty. The configuration does not escape physics; it declines a particular trade. What it pays instead is smaller, and — this is the part that matters for scaling — it does not grow faster than the aircraft.

5.5 A comparative sizing of three architectures

Supplementary S6 sizes three architectures against the same mission — this tail-sitter, a lift-plus-cruise aircraft, and a tilt-rotor — under three different sizing contracts: fixed fuel fraction, fixed fuel mass, and fixed maximum take-off mass with fixed payload. One set of equations serves all three, and every coefficient in it is back-solved from the light reference design of Section 6.2 rather than assumed. Mission, wing loading, disc loading, structural fraction and energy chain are held identical; only the cruise-drag multiplier and the architecture-specific mass differ. Under the first contract:

	Empty fraction	MTOW	Cruise L/D	Hover power	Range
A — tail-sitter	**0.580**	**50.0 kg**	12.00	**10.9 kW**	1 600 km
B — lift + cruise	0.689	86.0 kg	10.28	18.7 kW	1 370 km
C — tilt	0.624	60.3 kg	**13.44**	13.1 kW	**1 792 km**

The ordering depends on which contract is used, and that dependence is the result rather than an inconvenience. Range in the sizing equation contains the fuel *fraction*, so holding the fraction fixed lets the heavier aircraft carry proportionally more fuel and removes the mass bill from the range column altogether. Fixing the fuel *mass* makes range inversely proportional to take-off mass; fixing take-off mass and payload leaves fuel as the residual. These are three different questions, and the answers separate:

Range relative to the tail-sitter	Fixed fuel fraction	Fixed fuel mass	Fixed MTOW and payload
B — lift + cruise	−14.4 %	−36.5 %	−72.6 %
C — tilt	**+12.0 %**	+0.2 %	−19.1 %

Against lift-plus-cruise the conclusion is the same under every rule and grows more emphatic as the rule tightens, and the drag term driving it is a wind-tunnel result rather than an assumption. Against tilt it is not: of the twelve cells S6 reports, the tilting layout leads in three, all of them under the fixed fraction and all of them requiring its nacelles, pivots, actuators and hover-pitched blades to be credited as aerodynamically free. **No result from this section should be quoted without the rule it was computed under**, and no claim of superiority over the tilting family is made here in either direction.

What this comparison does and does not support. It supports the claim that the three bills are real and separable in a sizing loop. It does **not** support a claim that this configuration is better: the competing architectures are modelled from published mass fractions at a coarser level of detail than the one proposed here, which is modelled from a component build-up. **Comparing a build-up against a fraction favours whichever is modelled more optimistically**, and this study cannot rule out that it is this one. An external check against three flying eVTOLs, one per architecture, is reported in S6.1; it corroborates the ordering of the charges but is a comparison of other people's aircraft, not of this sizing. Section 8 states the comparison as conditional on both asymmetries.

6. Reference designs at two scales

A configuration argument is only as good as its willingness to become a number. This section sizes two aircraft from the arrangement of Section 4 — one at 50 kg and one at 1000 kg, a factor of twenty apart in mass — using the same equations, the same assumptions and the same architecture. The two points are not a light version and a heavy version of different aircraft. They are the same aircraft at two sizes, and the purpose of presenting both is to show that the proportions hold.

Every number below is calculated, not measured. Section 8 says what that means.

6.1 Sizing method

The method is elementary and the equations are given so that any result in this section can be checked by hand.

Hover. Thrust equals weight and induced power follows from momentum theory, $v_i = \sqrt{T/2\rho A}$ and $P_i = T^{1.5}/\sqrt{2\rho A}$, with disc loading $DL = T/A$ as the governing parameter; figure of merit gives shaft power.

Cruise. $C_L = W/(qS)$ fixes the lift coefficient at the chosen speed, $C_D = C_{D0} + C_L^2/(\pi ARe)$ gives the drag, and L/D is their ratio **at the cruise condition, not at the aircraft's best point**. The familiar $L/D_{max} = 0.5\sqrt{\pi ARe/C_{D0}}$ occurs at one particular speed — 25.3 m s^{-1} for the light design, only 1.26 times its stall speed. Cruising there would leave too little margin, so both designs cruise at 1.49 times stall and accept the lift-to-drag ratio that condition gives. Using the maximum value while quoting a different cruise speed would overstate the range, and an earlier version of this paper did exactly that.

Range. The series-hybrid chain is stated link by link rather than folded into one efficiency, because the result is sensitive to it and a reader should be able to disagree with any single link: engine 0.28, generator 0.90, power electronics 0.95, electric machine 0.92, propeller 0.80 — **overall 0.176**. Fuel energy is 12.9 kWh kg^{-1}. The engine figure matters most: 0.28 is representative of a small four-stroke at its best operating point, and it is why the range figures below are lower than an optimistic estimate would give.

Control. Tip-pair thrust follows from $M = 2TL = I\alpha$, with the transition manoeuvre as the sizing case.

Assumptions carried throughout: sea-level density; no compressibility; span efficiency $e = 0.85$; $C_{D0} = 0.0248$ for the light design, which is generous for a clean blended-wing body and absorbs the tip-frame contribution of Section 5.2 — that contribution is 0.0043, or seventeen percent of the assumed value, so the assumption is self-consistent rather than optimistic. Both are carried as assumptions throughout, so that every downstream figure rests on one stated basis. Section 6.6 computes both and reports what the computation does to them: it bounds them rather than replacing them, which is a weaker but more honest claim.

6.2 Light reference design — 50 kg

Quantity	Value
Maximum take-off mass	50 kg
Root chord	0.97 m
Tip chord	0.236 m
Span	3.45 m
Wing area	1.98 m²
Aspect ratio	6.03
Wing loading	25.3 kg m⁻²
Main propeller diameter	1.20 m
Disc loading	44.2 kg m⁻²
Tip propeller diameter	0.20 m
Frame post length	0.71 m each direction
Stall speed	20.1 m s⁻¹
Cruise speed	30 m s⁻¹ (108 km h⁻¹)
Cruise L/D	12.0
Hover power	10.9 kW
Cruise power, electrical	1.7 kW
Engine rating	2.6 kW
Battery buffer	1.8 kg (3.6 % MTOW)
Fuel	8 kg
Endurance	**14.8 h**
Range	**1 598 km**
Transition time	2 s

The mass budget behind this — 30 % structure, 16 % propulsion chain, 4 % battery, 8 % avionics and control, 16 % fuel, leaving 26 %, or 13 kg, for payload — is the allowance the design is sized against, and it is asserted here rather than derived. Section 6.7 rebuilds it from components and finds it can be met, with 2.2 kg in hand, on one condition that is not demonstrated: a structural areal density no greater than 1.78 kg m⁻². Paper aircraft are habitually lighter than the ones that get built, and no allowance for that has been paid in this table beyond the contingency inside the build-up. Section 8.2 keeps this as the single most likely place for these numbers to be wrong.

6.3 Heavy reference design — 1000 kg

Quantity	Value
Maximum take-off mass	1000 kg
Root chord	3.25 m
Span	11.55 m
Wing area	22.24 m²
Wing loading	45.0 kg m⁻²
Main propeller diameter	5.40 m
Disc loading	43.7 kg m⁻²
Tip propeller diameter	0.67 m
Frame post length	2.38 m each direction
Cruise speed	40 m s⁻¹ (144 km h⁻¹)
Cruise L/D	13.6
Hover power	216.2 kW
Cruise power, electrical	39.2 kW
Engine rating	54.3 kW
Battery buffer	40 kg (4.0 % MTOW)
Fuel	160 kg
Endurance	**12.6 h**
Range	**1 814 km**
Transition time	5.1 s

The heavy design has a longer range than the light one despite a shorter endurance. Both effects come from the same source: the larger aircraft cruises faster and, at a higher Reynolds number, achieves a lower zero-lift drag coefficient and therefore a better lift-to-drag ratio. Nothing in the architecture was changed to obtain this.

6.4 Scale behaviour

One qualification applies throughout: this is the scaling of the analytical sizing model — of powers, loadings and mass *fractions*. Whether the heavy design's structure closes depends on how shell areal density grows with size, which was not measured. Figure 11 shows the two designs at a common scale. Four properties are preserved and one is not.

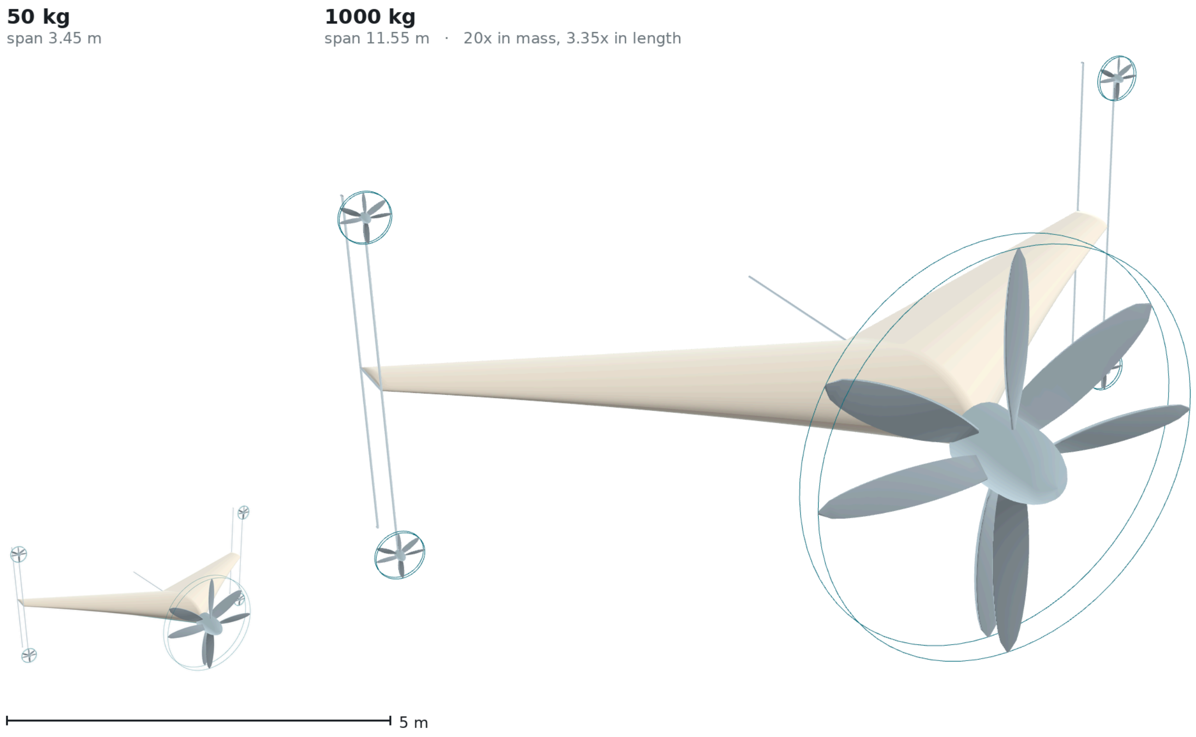

Figure 11. The two reference designs at a common scale: 50 kg and 1000 kg, twenty times apart in mass and 3.35 times in span. Disc loading, energy-buffer mass fraction and tip-frame drag fraction are preserved across that range; transition time is not, and neither is the ratio of propeller diameter to span, which rises from 0.35 to 0.47 because holding the disc loading makes the propeller grow faster than the airframe.

Disc loading is held constant — 44.2 and 43.7 kg m^{-2}. This is the rule that governs the sizing rather than a coincidence of it. Hover power per unit weight is $\sqrt{DL/2\rho}$, so fixing disc loading fixes specific hover power: hover power rises from 10.9 kW to 216.2 kW, a factor of 19.8 against a mass factor of 20. **Hover power grows linearly with mass rather than as the L^3.5 of the classical result**, and that is the whole benefit of fixing it. The cost is that disc area must then grow as L^3 rather than L^2, which for a fixed number of propellers is impossible. The architecture has two ways out and uses both: a coaxial pair may be added at no architectural cost, since every pair is torque-balanced on its own; and geometric similarity is not held.

The propeller therefore grows faster than the airframe. Wing loading rises from 25.3 to 45.0 kg m^{-2}, so span grows by 3.35 against the 4.50 by which the main propeller must grow. The ratio of propeller diameter to span rises from 0.35 to 0.47: the heavy design is not the light design photographed from further away, and its propeller occupies almost half its span. Nothing in the argument fails because of this — the propeller is the nose of the aircraft rather than an appendage, and disc loading is what is being held — but the claim that the configuration keeps its proportions applies to the quantities named here and not to every dimension. Much above 1000 kg, a single nose pair can no longer hold the disc loading and a second must be added.

The buffer fraction is preserved (3.6 % of MTOW at 50 kg, 4.0 % at 1000 kg), so the mechanism by which Bill 3 is avoided does not degrade with size, and **the frame drag**

fraction is preserved because frontal and wing area both scale as L^2.

Transition time does not scale, and this is the exception. The rotating moment follows $M = I\alpha$ with $I \propto mL^2$, so the moment needed to turn the aircraft in a fixed time grows much faster than the aircraft. Scaling the light design's two-second rotation to 1000 kg would demand 221.5 kW from the tip propellers — 102 % of hover power, which is to say it is not available:

Table 4. Tip-propeller power required to rotate the heavy reference design.

Rotation time	Tip-propeller power, 4 total	Fraction of hover power
2 s	221.5 kW	102 %
3 s	65.6 kW	30 %
4 s	27.7 kW	13 %
5.1 s	**13.4 kW**	**6 %**

The rule is that a larger aircraft turns more slowly. The heavy design rotates in 5.1 s at six percent of hover power — not a round number but the rotation time at which it holds the same control margin the light design holds at two seconds (Section 7.6). The constraint is less costly than it looks, because Section 7.4 shows a slower rotation loses *less* altitude: the scaling penalty on transition time works with the penalty on control power rather than against it. The classical objection to scaling a VTOL aircraft — hover power growing as L^3.5 against power available as L^3 — is removed on the hover side by fixing disc loading. It is not removed on the transition side, and Table 4 is where it reappears: **the rotation is the one place in this aircraft where the square–cube relation is still paid in full.**

6.5 Context

The following aircraft occupy the same mass range. They are listed to locate the reference designs in a real field, not to rank them.

Aircraft	MTOW	Payload	Payload fraction
HAVELSAN BAHA [8]	28 kg	2 kg	7.1 %
Textron Aerosonde Mk 4.7 VTOL [9]	45.4 kg	9.1 kg	20.0 %
Baykar KALKAN [10]	75 kg	~3 kg internal	4.0 %
HAVELSAN BULUT [11]	not published	5 kg	—
Elroy Air Chaparral [12]	865 kg	136 / 227 kg	15.7 / 26.2 %
Sabrewing Rhaegal-A [13]	1400 kg	360–450 kg	25.7–32.1 %
Pipistrel Nuuva V300 [14]	1700 kg	408 kg	24.0 %

All entries are from manufacturers' published material; payload definitions are not consistent between them and empty weights are generally not published.

Three statements can be made and a fourth cannot. The field is real and populated at both ends of the range. Payload fraction rises with size across it, from a few percent to roughly a quarter, which is the ordinary consequence of fixed costs not scaling down. And **none of these aircraft connects an internal-combustion engine directly to a lifting rotor** — every one uses either a generator or separate electric lift, which is independent confirmation that the series arrangement of Section 4.3 is the practical choice at this scale rather than an unusual one.

The fourth statement — that the reference designs outperform these aircraft — is not made. Sections 6.2 and 6.3 are calculated from a mass budget with an unpaid structural margin; this table describes aircraft that exist and fly. Placing a calculation beside a measurement and declaring a winner would be a category error. Several entries are also fully electric, for which endurance is set by battery specific energy rather than by configuration, so comparing them with a fuel-burning design would compare energy sources rather than architectures. What could properly be compared, once such aircraft are built, is **range at similar payload** — a configuration carrying a comparable load further is making an architectural claim, while one carrying a heavier load is making a claim about mass budgeting, which is the least validated part of this study.

6.6 Independent checks on the two assumed coefficients

Both coefficients carried through Sections 6.2 and 6.3 were assumed. Both have since been computed, and the computations are reported in full in Supplementary S1. Neither replaces its assumption in the figures above — those are quoted on one stated basis throughout — but each bounds it, and the direction of each is stated here.

Zero-lift drag. A Reynolds-averaged solution of the wing and body gives C_D0 between 0.0120 and 0.0148, against the 0.0248 assumed. The spread is the turbulence closure: 0.01475 with Spalart–Allmaras and 0.01201 to 0.01253 with k-ω SST, eighteen percent apart at matched wall resolution. **The assumption lies above the whole of that range**, so it is conservative rather than optimistic. Two things S1 does not settle: the solutions are fully turbulent, so the clean-surface figure of 0.0073 from the strip method is untested and the gap between it and 0.0120–0.0148 is now the largest single uncertainty in the zero-lift drag; and the wall-resolved SST case admits more than one stationary solution, two converged starts settling 4.3 percent apart in the pressure component.

Span efficiency. A vortex-lattice solution gives an inviscid span efficiency of 0.990 for the untwisted planform and 0.859 for the wing twisted to trim. Those are not the quantity the drag build-up needs: an Oswald-type efficiency also carries the viscous drag due to lift. An earlier version of this paper converted between them with a borrowed rule of 85 to 90 percent and reported an implied value of 0.735 to 0.78. S1 computes the conversion instead, on this planform, by calling the section solver at **each spanwise station's own local lift coefficient** rather than at zero lift and integrating the profile drag across the span — the two-dimensional-viscous-coupled-to-three-dimensional-circulation construction of the non-linear vortex-lattice literature [40]. Before any number is taken from it, the strip decomposition is checked against the solver it comes from: the strip loads reproduce the solver's own lift coefficient to six decimal places.

	Inviscid e	Oswald e	Ratio
Untwisted planform	0.990	0.931	0.940
Trimmed, −9° washout	**0.859**	**0.817**	**0.951**

The borrowed rule was wrong in the favourable direction and the conclusion is unchanged in the unfavourable one. The viscous penalty is 5 to 6 percent rather than 10 to 15, but the trimmed wing starts from 0.859, so the Oswald efficiency lands at **0.817 — below the assumed 0.85 by 3.9 percent**. At that value the cruise lift-to-drag ratio is **11.87 against 12.04**, and the range figures of Section 6.3 are optimistic by the same 1.4 percent. Two limits belong with the number: the vortex-lattice sections are symmetric, so a cambered section reaching the same local lift coefficient at lower incidence would carry less drag and the figure is a **lower bound**; and strip integration ignores sweep, which at 45° at the root is not a small omission, though the alternative — simple-sweep theory — halves the profile drag, which is a sign that the transformation does not apply to skin friction rather than a measure of the uncertainty.

What the vortex-lattice method is being asked for, and a bound on it that has no bound. Four results here come from a vortex-lattice solution: the span efficiency, the neutral point, the twist required to trim, and the roll damping. Falkner separates the quantities the method settles quickly — spanwise circulation, local aerodynamic centre — from those it does not, and every quantity taken here is of the first kind [24]. But a published comparison on a blended-wing-body of this class found the vortex-lattice lift coefficient low by **thirty to thirty-eight percent** against RANS, and excluded the method from its trim analysis on that basis [39]. The deviation there is nearly constant with incidence, which is the signature of a multiplicative error in the magnitude of the loading rather than an error in its distribution — and a factor common to lift and moment cancels in a ratio of derivatives. That argument depends on the moment scaling with the lift, and the pitching-moment comparison in that source is published with its values withheld. **The vortex-lattice results here therefore carry an untested magnitude error of unknown size, bounded above by a published comparison on a similar configuration, and every use made of them is of a kind a magnitude error *would not* disturb — provided the moment scales with the lift by the same factor, which is exactly what cannot be checked.** Section 8 lists settling this as the one exposure in the aerodynamic chain with no bound at all.

6.7 A component build-up of the mass budget

The sizing above assumes an empty-mass fraction rather than deriving one. Supplementary S2 builds the 50 kg design's mass item by item — structure from wetted area and an assumed shell areal density, tip frames sized by a vertical landing case, propulsion, energy, avionics and systems — and reports the break-even value of every assumption in it.

The build-up closes, and it closes on one number. The total is 11.88 kg against a 14.08 kg allowance, leaving 2.2 kg unallocated. It closes on the condition that the average structural areal density does not exceed **1.78 kg m^{-2}**, against the 1.5 assumed; at 1.78

the payload is gone. A doctoral study of unmanned-aircraft sizing reports 1.05 kg m^{-2} for a small UAV's monolithic composite fuselage shell supported by an internal structure [50], which places the assumption inside the range small composite airframes are built to without establishing that this airframe reaches it — that shell is carried by an internal structure, while this one is the wing and carries flight loads directly.

The build-up came in lighter than the target, and that is a warning rather than a result. Paper aircraft are habitually lighter than the aircraft that get built. What S2 does not contain is buckling, torsion, local load introduction, aeroelastic sizing, fasteners, adhesive, paint, or the mass of anything the design has not yet specified — and the 2.2 kg of margin is what all of those must fit into. Section 8.2 states the item as bounded from below rather than demonstrated.

7. Flight profile and transition

The transition between vertical and horizontal flight is the manoeuvre on which tail-sitters have historically been judged, and it is the part of this configuration that most deserves scrutiny. This section describes the flight profile, states the equations that govern the transition, and reports a simulation of it. One result contradicts a widely-assumed relationship and is presented as such.

7.1 The five phases

Stance. The aircraft rests on five points — the four lower ends of the tip frames and the aft end of the centre keel — with its longitudinal axis vertical. No launch equipment is present, and the aircraft is in its own storage attitude.

Vertical take-off. The nose pair spools to a thrust exceeding weight and the aircraft rises vertically. Attitude is held by the four tip pairs. This is the highest-power phase of the flight and the shortest.

Transition. The aircraft rotates from vertical to horizontal while accelerating, until the wing carries the weight. Treated in detail below.

Cruise. The aircraft flies as a tailless blended-wing body. The nose pair is now the cruise propulsor at its design point. The tip pairs provide pitch and yaw, and the lower-surface strip provides roll.

Landing. The reverse of transition, followed by a vertical descent onto the five contact points. Section 7.5 notes what is and is not analysed here.

Figure 9 shows the five phases in sequence. Nothing on the aircraft rotates relative to the aircraft at any point in it.

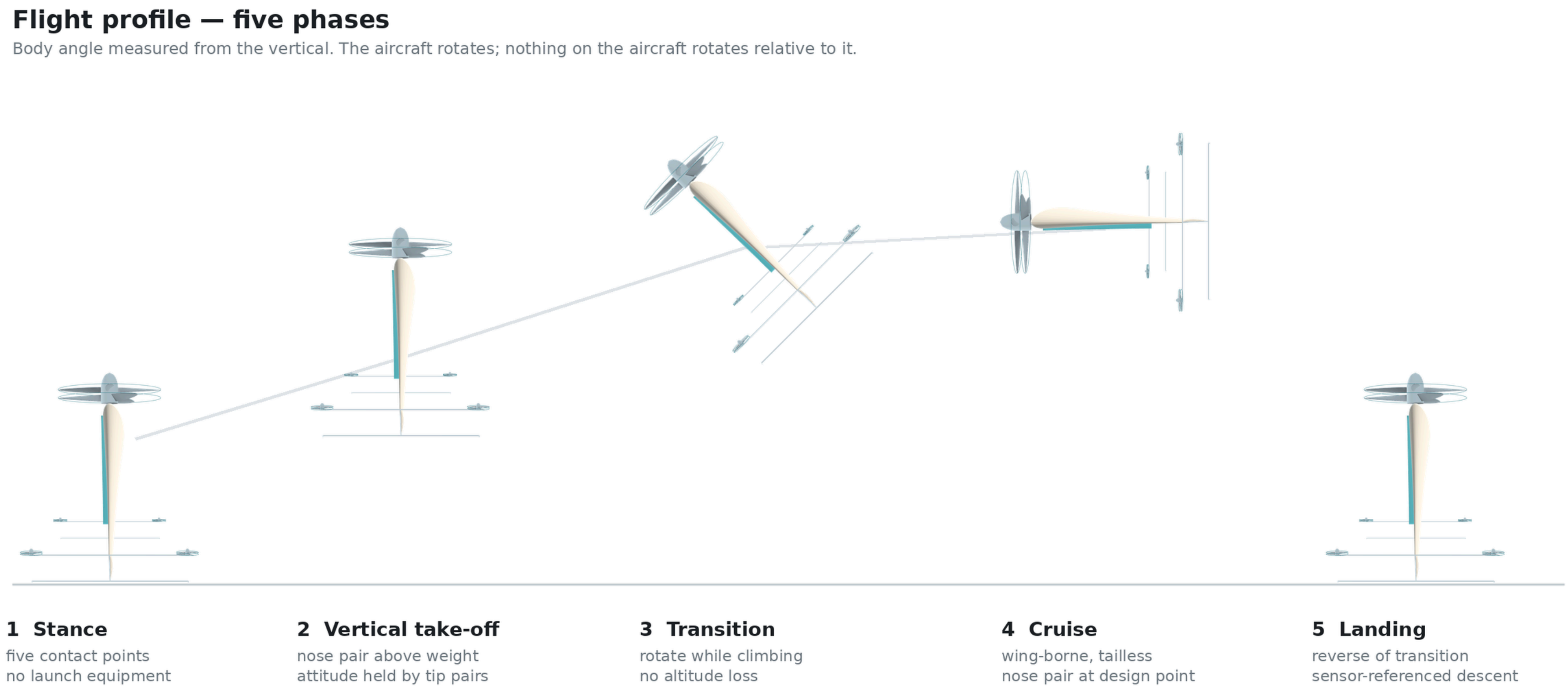

Figure 9. The five phases of the flight profile, drawn from the parametric geometry model. The rotation in phase 3 is treated in Sections 7.3 and 7.4.

7.2 Why the transition begins in the easiest condition

A common objection to tail-sitter transition is that the aircraft must fight the airflow while rotating. For a transition that begins in hover, it does not. At the start of the manoeuvre the airspeed is zero, so the free-stream dynamic pressure

$$q = \tfrac{1}{2}\rho V^2$$

is zero, and with it every aerodynamic moment that would resist the rotation. The aircraft is not turning against the air; it is turning in still air and then meeting the air as it accelerates.

The consequence is that the difficult part of the transition is not its beginning but its middle, where the airspeed has grown enough for aerodynamic moments to matter but the wing is not yet carrying the weight. The control authority requirement is set there, not at the start.

7.3 The thrust singularity that is never reached

Consider the aircraft at an angle θ from the vertical, and suppose for a moment that it must hold altitude with thrust alone. Vertical equilibrium then requires

$$T \cos\theta = W \quad \rightarrow \quad T = W / \cos\theta$$

which diverges as θ approaches ninety degrees. Read literally, this says a tail-sitter cannot complete a transition, and the expression is sometimes quoted to that effect.

The expression is correct and the conclusion drawn from it is not, because the premise is false. The aircraft does not hold altitude with thrust alone. Vertical support is

$$T \cos\theta + L = W, \qquad L = \tfrac{1}{2}\rho V^2 S\, C_L$$

and V is not zero during the rotation — it is growing, because the horizontal component T sin θ is accelerating the aircraft. The horizontal acceleration, in the same idealisation, is

$$a = g \tan\theta$$

so the very rotation that reduces the vertical component of thrust is what generates the airspeed that replaces it. The singularity is never approached because the wing arrives first.

This is the central mechanism of the manoeuvre, and it also explains the result of the next subsection.

7.4 Transition time: slower is better

The transition was simulated as a two-degree-of-freedom point mass. The body angle is driven from zero to ninety degrees over a rotation time t_r; thrust acts along the body axis, lift perpendicular to the velocity vector and drag opposite to it; the lift curve is linear to stall and a flat-plate relation beyond it. Altitude loss is the lowest point of the trajectory relative to the entry altitude. Figure 10a plots both reference designs at four thrust-to-weight ratios.

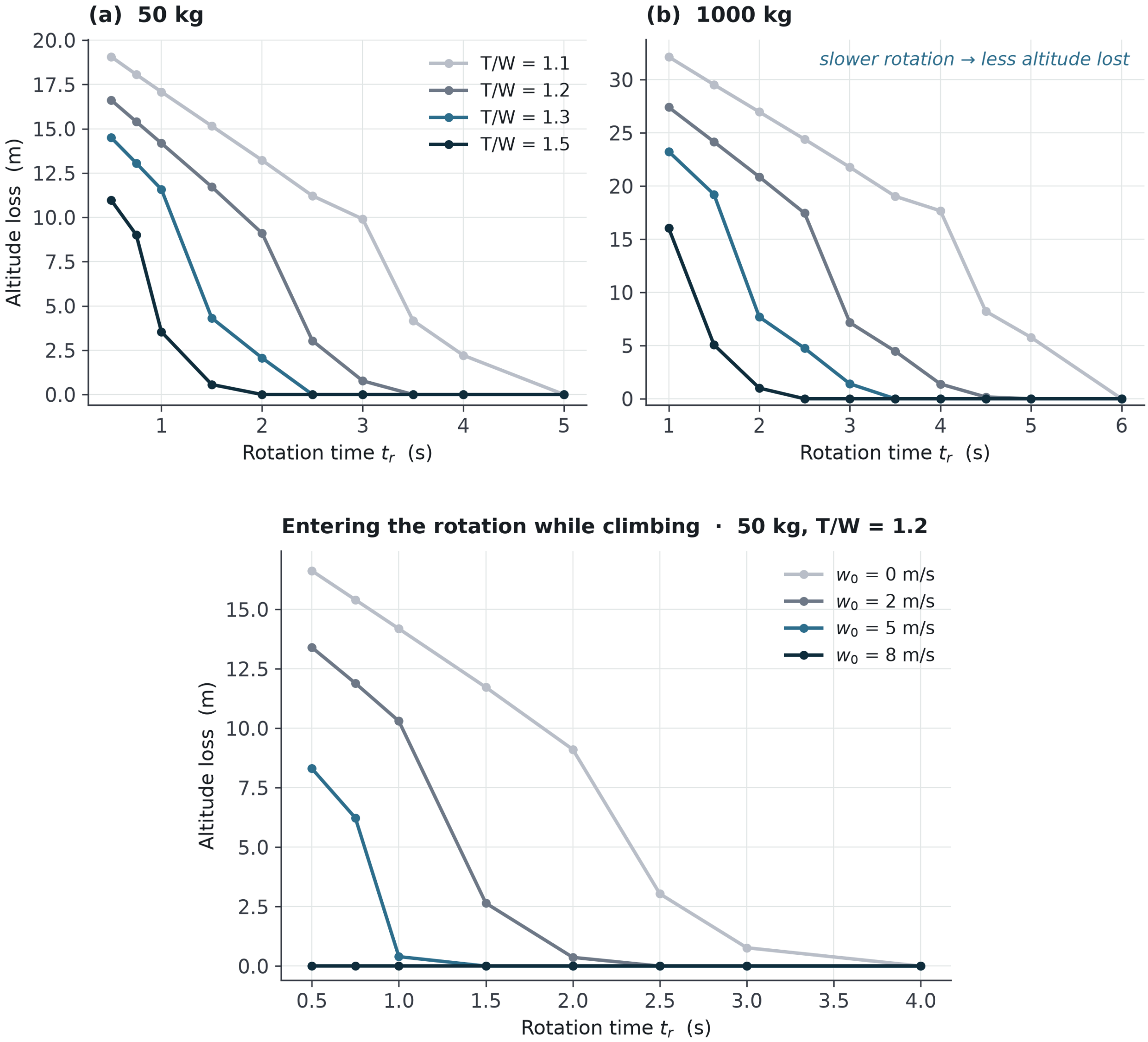

Figure 10. Transition altitude loss. (a) Both reference designs against rotation time at four thrust-to-weight ratios: a slower rotation loses less altitude, not more. (b) The light design at T/W = 1.2 entering the rotation with an upward velocity w_0; an entry climb of 5 m s^{-1} removes the loss at every rotation time considered.

The ratio the aircraft actually has must be established first, and it is not a free choice. Section 6.1 sizes hover power at thrust equal to weight, so the 10.9 kW of Section 6.2 buys T/W = 1.00 and nothing more; the same is true of the 216.2 kW of Section 6.3. The only other source of vertical thrust on this aircraft is the tip pairs, and during the rotation they are occupied producing the rotation itself. On the bang-bang profile the upper pairs run at full thrust and the lower pairs at zero, which is the M = 2TL of Section 4.3 — and the two upper pairs still push upward. That fixes the ratio available *during* a full-authority rotation at **1.066 for the light design and 1.041 for the heavy one**, and it is the ratio the tables below use. Giving up rotation authority buys a little more, to 1.132 and 1.082 with none retained; Supplementary S2 gives the trade. An earlier version of this section assumed T/W = 1.2, which the installed power does not supply at any setting, and the tables have been recomputed.

t_r	Light, 50 kg	t_r	Heavy, 1000 kg
1 s	−18.2 m	2 s	−31.0 m
2 s	−14.7 m	3 s	−26.9 m
3 s	−11.2 m	4 s	−22.7 m
4 s	−4.9 m	5.1 s	−13.1 m

The relationship is monotonic in the direction opposite to the one usually assumed. It is frequently supposed that a tail-sitter should rotate as fast as possible, on the reasoning that it is unsupported during the rotation and therefore falls for a time t_r, giving a loss proportional to t_r^2. **That reasoning is wrong, and the error is in its premise:** the aircraft is not unsupported. Vertical support is T cos θ + L, and a slow rotation keeps cos θ large during exactly the interval in which speed, and therefore lift, is being built. A fast rotation collapses cos θ before there is any lift to replace it, and the aircraft falls precisely because it hurried.

The practical consequence is a simplification rather than a trade. The control *moment* required to rotate in time t_r scales as $1/t_r^2$ and the control *power* as $1/t_r^3$ — Table 4 of Section 6.4 is the second of these, and its entries are constant to within a third of a percent when multiplied by t_r^3 — so a slow rotation is cheap in authority and cheaper still in power; and altitude loss also falls with t_r. **All of these point the same way**, so there is no optimum transition time to be found between competing penalties — the rotation time is set by what the actuator can do, not by a balance, and Section 7.6 shows that is where both reference times come from.

Entering the rotation while still climbing removes the penalty entirely, and it survives the correction to thrust-to-weight above. At an entry climb of 5 m s^{-1} the altitude loss is zero at both reference rotation times — 2 s light and 5.1 s heavy — and remains zero at every ratio from 1.066 down to 1.00, which is to say the result does not depend on the tip pairs contributing any lift at all once the climb has been acquired.

Acquiring the climb is where the correction is paid. The aircraft reaches transition altitude by climbing, so it need not stop and hover first, but the excess thrust available to build that climb is now 0.132 g rather than the 0.2 g an earlier version claimed, and only if no rotation authority is held in reserve; with full authority retained it is 0.066 g. Five metres per second is therefore reached in 3.9 s over 9.6 m at best, and 7.7 s over 19.3 m at worst, against the 2.6 s and 6.4 m previously stated. The energy involved is unchanged and remains negligible — 625 J against a fuel energy of 103 kWh — so what the correction costs is time and height, not range. **The reference profile is therefore still to enter the rotation at 5 m s^{-1} of climb**, with the acquisition charged at the achievable rate.

Two consequences follow that the earlier tables hid. Starting the rotation from rest is worse than reported — the light design loses 14.7 m at its own two seconds rather than 9.1 m, and the heavy design 13.1 m at 5.1 s rather than none — so the climb entry is not a convenience but a requirement. And the tip pairs, introduced in Section 4.3 as moment producers and charged in Section 5.4 for their mass and drag, turn out to carry the take-off thrust margin as well: an aircraft whose primary propulsor is sized at thrust equal to weight leaves the ground on them. That is a second duty for hardware bought for the first, which is the kind of economy this configuration is built on — but it is also a dependency, and it is a harder one than it looks, because the margin and the attitude authority are drawn from the same four propellers and cannot both be had in full. Section 8 records it as an open item.

The test is a lower bound. A point mass carries no rotational dynamics, no aerodynamic pitching moment and no control-power limit; Section 7.6 supplies the rotational budget that this model omits, and Section 8 states what neither supplies.

7.5 Landing

Landing reverses the sequence: the aircraft decelerates, rotates nose-up, and descends vertically onto its five contact points. Two things should be said about it plainly.

The first is that the historical objection to this manoeuvre does not apply. The XFY-1 was cancelled because its pilot had to judge a backwards vertical descent by looking over his shoulder. There is no pilot here, and height above ground is a sensor measurement rather than a human estimate.

The second is that the vertical descent itself has not been analysed in this study. A rotor descending into its own wake can enter the vortex ring state, in which thrust becomes erratic and increasing power makes matters worse. Whether the descent profile of this configuration enters that region, and at what rate of descent, is an open question. It is listed in Section 8 rather than answered here.

7.6 Whether there is enough authority to rotate, and whether it trims

Rotating the airframe through ninety degrees is the manoeuvre this configuration must perform with four small propellers and no control surfaces. Supplementary S4 carries the full budget — inertia derivation, rotation profiles, centre-of-gravity window, twist sweep and the measured section evidence. This section states what it returns.

The rotation closes at the actuator limit rather than clear of it. The pitch inertia derived from the component build-up is 9.81 kg·m^2 for the light design and 2 503 kg·m^2 for the heavy one. On the cheapest rotation profile the tip propellers carry the manoeuvre with a margin of **1.49** at the light design point and **1.57** at the heavy one; on a smoothly commanded profile the margins fall to 0.99 and 1.05. The reference rotation times — two seconds light, 5.1 seconds heavy — are therefore lower bounds set by the actuator, not comfortable choices, and the margin narrows with size.

The light figure rests on a tip thrust the design tables assert rather than derive, and on the conservative basis it is thinner still. The 16.2 N quoted per pair implies a figure of merit of 0.702, against the 0.599 used for hover everywhere else. Recomputing at 0.599 with a fifteen percent coaxial interference loss gives 12.4 N, an available moment of 17.6 N·m, and margins of **1.14 bang-bang and 0.76 smooth** — which is to say the light design closes on the cheapest profile and does not close on a smooth one at two seconds. The heavy design was computed on the conservative basis from the outset. Supplementary S4 gives both, and the honest reading is that the rotation is sized by the actuator under either basis and has no margin to give under the stricter one.

Resolving the requirement along the trajectory changed the question rather than merely quantifying it. The aircraft does not reach ninety degrees of incidence: the body rotates through ninety, but the relative wind rotates with it, and **peak incidence is 17.5° for the light design entering in a 5 m s^{-1} climb and 21.6° entering from level hover**. The high-incidence part happens at low dynamic pressure, where the margin tolerates a pitching-moment coefficient of 0.205; the tight part is the *end* of the rotation, where incidence is small and speed is high, and the budget there is 0.050. **The demanding case is not the post-stall middle but the attached-flow end**, which makes it a trim question rather than a stall question.

One mechanism was omitted and including it improves the case. The inboard half of the wing lies in the nose propeller's slipstream, where the local flow is faster and more axial, so the effective incidence there is lower than the geometric one. Applying the slipstream relation used in the tail-sitter literature [23] gives **four to eight degrees effective against seventeen to twenty-two geometric**: half the wing is not post-stall at the moment of peak incidence. The measurement still owed concerns the outboard half.

Static stability is shown, and the trim chain closes by twist. A vortex-lattice solution places the neutral point at 0.859 m from the root leading edge — 34.4 percent of mean aerodynamic chord — and the packaging centre of gravity at 80.2 percent of root chord gives a **static margin of +12.5 percent of mean aerodynamic chord**, with a pitching moment of 0.056 to be balanced at the cruise lift coefficient. Two published benchmarks place that window favourably: a blended-wing UAV of this class reports its own margin of 0.081 as "marginally outside the typical range for static longitudinal stability", given as 0.1 to 0.3, and its $C_{m\alpha}$ of −0.086 per radian against a typical −0.3 to −1.5 [41]. This configuration sits inside both at 0.125 and −0.48 per radian.

Reflex does not supply the 0.056, and this is now a measured statement rather than an inference. Nine reflexed and low-moment sections have been tested in tunnels that measure pitching moment. Exactly one returns a positive value — NACA 2R212 at **+0.004** [17], one fourteenth of what is needed. The three 1930s reflexed sections whose mean lines were shaped from thin-aerofoil theory to give *zero* quarter-chord moment measure "practically zero", from −0.001 to −0.007 [45]; the four sections of the NACA 4400R family were designed to a target of **−0.03** and the report states that "the design pitching-moment coefficient was realized" [46]. **Reflex, as actually built and measured,**

is a device for removing negative pitching moment rather than for producing positive pitching moment. Two costs are measured with it, and both bear on a tail-sitter: maximum lift falls by about twelve percent in the first family and ten percent in the second — and maximum lift is what a tail-sitter needs at the high-incidence end of transition. A flying tail-sitter shows where the positive moment actually comes from: it uses a symmetric section, and "the upward trim of elevons makes the symmetric airfoil to have reflexed camber line", producing the positive moment at the aerodynamic centre [51]. The reflex that trims that aircraft is a deflected control surface held permanently out of line, not a property of its section.

Nine degrees of tip washout trims this aircraft at cruise with no camber at all, and the vortex-lattice model computes it directly because the mechanism is geometric rather than sectional: on a swept wing the tips lie well aft, so negative tip incidence produces a nose-up moment about the centre of gravity. The price is a span efficiency of 0.865 instead of 0.993 and a cruise lift-to-drag ratio of 12.11 instead of 12.65 — **4.3 percent of cruise efficiency, paid to be tailless**, and entered in the ledger of Section 5.4 as its fifth item. The reflex route is not free either: a blended-wing UAV trimming by reflex rather than twist records that carrying reflex over a wide span "is not conducive to the improvement of overall lift-to-drag performance" [44]. **Both roads to trim on a tailless configuration cost cruise efficiency**, which is the reading Section 5.4 places on the 4.3 percent: it is the price of having no tail, not the price of choosing the wrong way to do without one.

The twist earns its cost three times over. It closes the trim chain; on a swept planform it delays tip stall, which on a tailless aircraft matters more than usual because a tip stall moves the centre of pressure forward and there is no tail with which to argue; and it inverts the stall sequence behind a blended-wing pitch-break. That third mechanism was found late and does not depend on aspect ratio: in a blended-wing-body analysed by both a low-fidelity method and RANS, the moment prediction departs above eight degrees because "the main wing stalls before the main body, causing the BWB to pitch-up" [39]. Washout makes the root stall first, and on a blended wing the root is the body.

The pitching moment through transition remains this study's largest open item, and the reason nobody computes it has been measured. A small blended-wing-body UAV was analysed by RANS and then tested in a wind tunnel at a Reynolds number of 2.0×10^6: from $-6°$ to $10°$ of incidence "both the aerodynamic force values and the variation trends are in quite good agreement"; from $10°$ to $26°$ they "show remarkable differences between the numerical and experimental results" [44]. The same boundary appears at lower fidelity — a vortex-lattice solution of this class of configuration departs above eight degrees [39] — and the incidences this aircraft passes through lie inside that band. **Three methods of three fidelities fail at the same place, and the highest of them fails against measurement.** The item therefore belongs to measurement rather than to computation, and Section 8 asks for it as measurement. The same tests found the blended-wing configuration to have "soft-stall performance", which is the benign end of the range the pitch-up literature describes.

The field does not have this term either. A transition-optimisation study with outdoor flight trials carries **no pitching-moment term at all** [28]; a second carries a linear one and flight-tested the question this configuration asks, reporting that without elevons its tail-sitter had "a well-controlled attitude response during hovering and transition" but that manoeuvres in level flight caused "an oscillatory attitude response" with "motor saturations observed", attributed to "the increased aerodynamic moment but decreased motor thrust at high-speed level flight" [29]. A survey records the same outcome for a separate vehicle: "able to achieve transition to forward flight, but they had poor control over the vehicle once in forward flight" [30]. A third carries a nonlinear $C_m(\alpha)$ but borrows the curve and closes the remaining discrepancy with an adaptive law [42]. **Every one of them obtains the transition aerodynamics by borrowing, fitting or adapting, and none by measuring the vehicle it flies.** Two independent programmes, different vehicles, the same division — transition passed, forward flight difficult — and that is the same structural tension this section derives from the moment budget: the tight case is the end of the rotation and beyond, where aerodynamic moment grows as V^2 while propeller thrust falls.

8. Limitations

This is a configuration study. It contains no experimental validation of any kind, and the numbers in it are the output of elementary methods applied to a set of assumptions. This section states what those limits are, in enough detail that a reader can judge how much weight each result will bear. Several of the items below were discovered during the study and changed its results; they are recorded here rather than smoothed away.

8.1 What is not shown

No part of this study has been validated experimentally. No wind-tunnel test, no flight test, no hardware. Everything here is calculation, and the calculations rest on assumptions stated in the sections that use them. Supplementary S5 enumerates every limitation item by item with the break-even value of each assumption that has one. This section states the ones that bear on the conclusions.

8.2 The three that could change a conclusion

The buffer's specific power is the most exposed number in the paper. Bill 3 is avoided by sizing the engine for cruise and supplying the hover excess from a battery buffer, and the light design's buffer implies **5.63 kW kg^{-1}** to hover and **6.48 kW kg^{-1}** to leave the ground. Both figures are taken at the electrical bus, where the buffer is; an earlier version of this paper differenced the rotor shaft against the engine shaft and reported 4.61, which understated the demand by a fifth. A 24S nickel–cobalt–manganese pack designed, built, bench-tested from 0.2 C to 10.68 C and flown in an electric VTOL aircraft measures **724 W kg^{-1}** continuous for the unit pack and **892 W kg^{-1}** for the flight system, reaching roughly 1.5 kW kg^{-1} at its maximum tested rate with a thermal margin of 4.9 °C [47]. A NASA-funded design study adopts 4 kW kg^{-1} and states that this is "about twice that of existing batteries" [48]. Sized at the measured thermal ceiling the buffer is

6.8 kg rather than 1.8 kg, against 2.2 kg of unallocated mass; sized at the measured continuous figure, 11.4 kg, which is nearly the whole payload. **The light design's mass budget does not close at any measured specific power.** The defence available earlier — that a short-duration buffer is a different product from an energy-optimised automotive pack — does not survive, because the source above *is* that product. The gap to be closed is a factor of 3.8 on the measured ceiling and 6.3 on the measured continuous rate. What would resolve it is a pack demonstrating that at acceptable temperature, or a heavier buffer carried at the cost of payload fraction.

The aircraft leaves the ground on its control propellers, and that is a dependency rather than a design feature. Section 6.1 sizes hover power at thrust equal to weight, so the primary propulsor supplies T/W = 1.00 exactly and no more. The margin to take off comes from the four tip pairs, which raises the achievable ratio to 1.066 with full rotation authority retained and 1.132 with none — never to the 1.2 an earlier version of Section 7.4 assumed. Three things follow and none of them is closed here: the take-off margin and the attitude authority are drawn from the same four propellers and compete; the buffer must carry the tip pairs as well, which is the 6.48 kW kg^{-1} above; and a rotation entered from rest, rather than from a climb, now loses 14.7 m at the light design's own two seconds. The entry climb of Section 7.4 is therefore a requirement of the configuration and not a refinement of it.

The mass budget is bounded from below and the bound is thin. The component build-up of Supplementary S2 totals 11.88 kg against a 14.08 kg allowance, on the condition that average structural areal density does not exceed **1.78 kg m^{-2}** against the 1.5 assumed; at 1.78 the payload is gone. A doctoral study of UAV sizing reports 1.05 kg m^{-2} for a small UAV's monolithic composite fuselage shell supported by internal structure [50], which places the assumption inside the range small composite airframes are built to without establishing that this airframe reaches it — that shell is carried by internal structure, while this one is the wing and carries flight loads directly. The build-up contains no buckling, torsion, local load introduction, aeroelastic sizing, fasteners, adhesive or paint, and 2.2 kg is what all of those must fit into. **Paper aircraft are habitually lighter than the aircraft that get built.**

The transition pitching moment is not computed, and Section 7.6 shows the computation is not currently reliable for anyone on this class of configuration. Three methods of three fidelities fail above roughly ten degrees of incidence, the highest of them against wind-tunnel measurement [39,44], and the incidences this aircraft passes through lie inside that band. **This item belongs to measurement.** It asks for the pitching moment to about twenty-two degrees at low dynamic pressure, on the outboard half of the wing — the inboard half lies in the slipstream and sees four to eight degrees — and for trim at cruise incidence.

8.3 What is assumed rather than derived

The planform's sweep, taper and thickness distributions were **chosen, not optimised**. The centre of gravity is a packaging assumption. The zero-lift drag coefficient of 0.0248 and the span efficiency of 0.85 are assumptions; Section 6.6 bounds both by calculation and neither replaces its assumption — the span efficiency computes to **0.817, optimistic by 3.9 percent**, worth 1.4 percent of cruise lift-to-drag ratio and of the ranges quoted. Torque balance is exact at cruise only, leaving a small residual in hover. The comparative sizing of Section 5.5 is conditional on the two competing architectures being modelled at the same level of detail as this one, which they are not: they are modelled from published fractions.

The vortex-lattice results carry an untested magnitude error. A published comparison on a blended-wing-body of this class found the vortex-lattice lift coefficient low by thirty to thirty-eight percent against RANS [39]. Section 6.6 argues that a near-constant multiplicative error of that kind cancels in the ratios this paper takes from the solution — neutral point, static margin, twist effectiveness — but the check that would confirm it is withheld in that source. **This is the one exposure in the aerodynamic chain with no bound at all.**

8.4 What is sized but not closed

Attitude control is sized in every axis and closed in none. Roll asks for **$\Delta C_L \approx 0.12$** from the strip, which published fence and Gurney data make plausible without establishing it for this geometry, and the strip's measured weaknesses — degradation in turbulence, contested post-stall behaviour, a dead band below seven percent of travel — all fall in conditions this configuration uses it in. Directional stability asks for a **39 mm** fairing chord on a frame that already exists, at a Reynolds number near 80 000 where thin symmetric sections are measured to be nonlinear about zero incidence; whether such a fairing develops the side force credited to it is not established, and the toe-out its aspect ratio requires carries a stall-related failure mode at large sideslip that has not been computed. Hover disturbance rejection, ground handling, crosswind and vertical descent have been checked only to order of magnitude or not at all.

8.5 Where this could most efficiently be attacked

Supplementary S5 lists six places. Two have been carried out and are folded into Section 6.6: a three-dimensional solution for the centre body, and a viscous solution of the twisted planform station by station. Of the remaining four, the one with no bound at all is a Reynolds-averaged or panel solution of this planform's loading, to bound the magnitude question above. **Three of the four can be carried out computationally; the fourth cannot, and saying otherwise was the most consequential thing this study got wrong about itself.** Transition controllability rests on a pitching moment that three methods of three different fidelities fail to predict above roughly ten degrees of incidence, the highest of them against wind-tunnel measurement, so it is blocked on data rather than on effort — as is the fin derivative of Section 4.4, for the reason Supplementary S3 gives. An earlier version of this section stated that none of the four required an experiment. The configuration is described in enough detail in Sections 4 and 6 for another group to attempt any of them independently, by whichever of the two routes each one needs. The computational setup, the grid-convergence study and the record of what failed along the way are in the repository, so every result here can be re-run and checked rather than taken on trust.

9. Conclusion

Hybrid vertical take-off aircraft pay for their vertical capability, and the payment is architectural rather than a defect of implementation. It appears in three currencies — the mass of hardware carried but unused, the drag of hardware exposed but inactive, and a power system sized by a condition that holds for about two percent of the flight — and **each of the architectural moves surveyed here reduces one of them by increasing another**. They are not offered as the only costs a VTOL aircraft carries; they are the three that follow from the duty-cycle mismatch, and the claim is about them. A NASA sizing study of four VTOL architectures reaches the same conclusion from the opposite direction, finding the lift-plus-cruise concept the heaviest of those examined while also the most efficient in cruise, and naming the cause as the empty weight carried for hover.

Stating the tax that way makes its escape condition explicit, and it has four parts: **it is charged unless hover and cruise are served by the same hardware, doing the same job, in the same orientation, with the hover peak drawn from a buffer rather than from permanently installed continuous power.** The configuration described here satisfies that condition rather than compensating for failing it. The aircraft rotates; nothing on the aircraft rotates relative to it. A single coaxial pair at the nose provides all thrust in both regimes; four small coaxial pairs at the tips provide moments and nothing else; and a strip on the lower surface is assigned the one gap propellers cannot close — the rolling moment, which parallel thrust vectors cannot produce at any setting or mounting position. There are no elevons, no rudder, no tilting mechanism, no retraction mechanism and no dedicated lift system.

The configuration was sized at 50 kg and at 1000 kg with the same equations, twenty times apart in mass. Disc loading is constant by design, which is what keeps hover power growing linearly with mass instead of as the classical L^3.5; the buffer that decouples the engine from the hover peak stays under four percent of take-off mass at both points; and the drag fraction charged to the tip frames is preserved. Three things do not scale, and all three are reported rather than smoothed: the larger aircraft must rotate more slowly; its propeller grows faster than its span, so the heavy design is not the light design seen from further away; and these are properties of the sizing rules, while a component build-up of the structure meets the mass fractions at 50 kg and does not at 1000 kg.

Two results emerged during the study that changed it. The tip frames, left as circular tubing, would produce nearly as much drag as the rest of the aircraft — fairing them is a requirement rather than an option, and the same fairing turns out to be the aircraft's directional stability surface. And **a slower rotation loses *less* altitude, not more**, because the aircraft is supported during the manoeuvre rather than falling through it, so entering the rotation while still climbing removes the altitude penalty entirely in the point-mass model. That second result survived a correction that might have removed it. The primary propulsor is sized at thrust equal to weight and therefore supplies no climb at all; the margin comes from the four tip propellers, which raises the achievable ratio to between 1.066 and 1.132 rather than the 1.2 an earlier version assumed. Recomputed there, the altitude loss at both reference rotation times is still zero — but acquiring the entry climb now takes twice as long, a rotation begun from rest is worse than reported, and the take-off margin and the attitude authority are drawn from the same four propellers and compete for them.

What this paper offers is a configuration and its numbers, not a validated aircraft. There is no wind-tunnel data here and no flight test. Of the analyses Section 8 lists as tests of these results, three have been carried out. A three-dimensional solution for the centre body narrowed the zero-lift drag without overturning the assumption. A component build-up closes the light design with 2.2 kg in hand, conditional on a shell areal density at or below 1.78 kg m^{-2} *and* on a battery buffer no measured cell can yet supply, and does not close the heavy design. And a rotational check shows the tip propellers can turn the aircraft's own inertia through the transition with a margin of 1.49 at 50 kg and 1.57 at 1000 kg on the cheapest profile — **0.99 and 1.05 on a smooth one**, and 1.14 and 0.76 at 50 kg if the tip thrust is recomputed on the figure of merit used elsewhere in the paper rather than the one its design table implies. Under every basis, both reference rotation times are actuator-limited lower bounds rather than comfortable choices.

Resolving that check along the trajectory changed the question. The aircraft does not reach ninety degrees of incidence: the body rotates through ninety but the relative wind rotates with it, and peak incidence is seventeen to twenty-two degrees — four to eight over the half of the wing lying in the nose propeller's slipstream. **The tight case is not the high-incidence middle but the end of the rotation**, where incidence is small and speed is high, which makes it a trim question rather than a stall one. The configuration is statically stable, with a neutral point at 34 percent of mean aerodynamic chord and a margin of 12.5 percent, and the moment to be balanced at cruise is 0.056. Reflex does not supply it: nine reflexed and low-moment sections have been measured in tunnels that measure moment, exactly one returns a positive value — one fourteenth of what is needed [17] — and the rest lie between zero and −0.03, because reflex as actually built is a device for removing negative pitching moment rather than producing positive moment [45,46]. **Nine degrees of tip washout does supply it**, at a cost of 4.3 percent of cruise efficiency: the price of having no tail.

That twist also settled an assumption, and not in the paper's favour. Computing the profile drag of the trimmed wing station by station at each station's own local lift coefficient gives an Oswald span efficiency of **0.817** against the 0.85 assumed — optimistic by 3.9 percent, worth 1.4 percent of the ranges quoted.

Attitude control is sized in every axis and closed in none, and that is the honest summary of its control case. Roll asks for $\Delta C_L \approx 0.12$ from the strip, which published fence and Gurney data make plausible without establishing it here. Yaw is the strongest axis the arrangement has, at 2.4 times the pitch moment, because differential thrust between the left and right tip pairs acts through the semi-span; what yaw lacks is not authority but stability, and the surfaces that must supply it are the same tip-frame fairings.

That one member serves as landing structure, moment arm, propeller mount and directional stability surface is the configuration's own argument made once more; that none of the four roles has been verified together is its principal limitation.

An earlier version of this section stated that none of the remaining analyses required an experiment. **That is no longer true, and the change is the most important thing this study learned about itself.** Transition controllability rests on a pitching moment that three methods of three different fidelities fail to predict above roughly ten degrees of incidence — the highest of them against wind-tunnel measurement — and the paper declines to substitute a reduced calculation for a measurement.

What survives independently of that is the framework. The three currencies, the demonstration that architectural remedies transfer the penalty rather than remove it, the escape condition, and the finding that architectural comparisons change their ranking with the sizing contract chosen — none of these depends on whether this particular aircraft is ever built. meryemAircraft is the case that shows the escape condition can be instantiated in a real geometry and carried through to reference designs at two scales. It is not offered as a validated vehicle, and the paper is careful throughout to say which of its statements are demonstrated, which are conditional, and which are open. The configuration is described in enough detail for another group to attempt any of the outstanding analyses independently, and that is the outcome this paper is written to invite.

Declarations

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Conflicts of interest. The authors have filed a patent application covering the aircraft configuration described in this paper (Türkpatent application 2026/014570).

Data availability. All data supporting the reported results are contained within the article. The parametric geometry model, the figure-generation scripts and the transition simulation, together with the aerodynamic calculations of Section 6.6 — including the mesh generator, the case setup, the grid-convergence study and the wall-resolution and turbulence-model sensitivity runs behind the computed zero-lift drag — are openly available at https://github.com/LORDTEK/meryemAircraft

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